SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): UNDP Ethiopia
Solution Maker (SM): Kitovu Technology

1. Challenge Definition

1.1 Refined Challenge Statement

Ethiopia is rapidly developing its national framework for climate action and carbon markets, yet it faces a critical challenge in ensuring robust, transparent Monitoring, Reporting, and Verification (MRV) of emission reductions. The current manual and fragmented MRV processes struggle to keep pace with the country's ambitious climate initiatives, risking data gaps, higher costs, and limited trust among stakeholders. Kijani MRV addresses this gap by introducing a **digital verification layer**. This blockchain-backed platform will immutably track carbon project data to **complement** (not replace) Ethiopia's national MRV system. This refined challenge focuses on building **trust, traceability, and transparency** in carbon credit generation for Ethiopia's emerging carbon market. By piloting Kijani MRV in real projects, we aim to demonstrate how digital innovation can **streamline carbon accounting**, enforce high-integrity standards, and empower local stakeholders, all while aligning with national policies and UN Sustainable Development Goals.

1.2 Local Context

Ethiopia is one of the most climate-vulnerable countries globally, where agriculture, forestry, and household energy use are deeply intertwined with rural livelihoods and economic stability. More than **65 percent of the population depends on agriculture**, while **over 85 percent of households rely on biomass fuels**[1], primarily firewood and charcoal for cooking[2]. This heavy reliance on natural resources has accelerated environmental degradation as Ethiopia continues to lose an estimated **92,000 hectares of forest annually**[3], with forest cover declining to approximately **15.7 percent of total land area in 2023**[4]. Rural poverty remains widespread, affecting over **30 percent of the population**[5], and recurrent climate shocks, droughts, floods, and land degradation frequently reverse hard-won development gains.

These structural challenges are directly linked to Ethiopia's climate ambitions. The country faces an estimated **USD 316 billion financing gap** to achieve its Nationally Determined Contribution (NDC) targets by 2030[6]. At the same time, global demand for **high-integrity carbon credits is projected to**

---

[1] **International Energy Agency (IEA)** & **Clean Cooking Alliance** (2023).
[2] **World Bank (2023).** *Ethiopia Country Climate and Development Report (CCDR).*
[3] **FAO (2022).** *FAOSTAT Forestry Database – Ethiopia.*
[4] **Global Forest Watch (2023).** *Ethiopia Forest Cover and Tree Loss Data.*
[5] **Central Statistical Agency (CSA) of Ethiopia** & **World Bank** (2022–2023).
[6] **UNDP (2023).** *Climate Promise: Ethiopia Financing Needs Assessment.*

reach USD 250 billion annually by 2030[7], creating a major opportunity for countries like Ethiopia that possess significant mitigation potential across forestry, agriculture, and clean energy[8]. However, without **trusted, transparent, and scalable systems for Monitoring, Reporting, and Verification (MRV)** and without a credible digital carbon registry, Ethiopia risks being unable to fully access or compete in these emerging carbon markets.

The communities most affected by these gaps are **smallholder farmers, women, and youth in rural areas**, who depend directly on forests and land for survival. These groups also stand to benefit the most from carbon revenues that can finance forest restoration, deforestation-free value chains, clean cooking technologies, and climate-smart agriculture, if such revenues can be verified, trusted, and fairly distributed.

In response, Ethiopia has taken important policy and programmatic steps. The government enacted a **Carbon Market Regulation in 2024**[9], signaling its intention to participate in international carbon markets and prepare for Article 6 cooperation under the Paris Agreement. Large-scale initiatives such as the **Green Legacy Initiative**, which has reportedly planted over **32 billion seedlings since 2019**, demonstrate strong political commitment to restoration and climate action[10]. Parallel efforts in clean cooking implemented through programmes such as **Ci-Dev, World Vision**[11]**, and newer market-based approaches** aim to reduce biomass dependence, improve health outcomes, and curb deforestation.

However, despite these advances, a critical gap remains, as many initiatives continue to rely on ***fragmented and predominantly offline MRV and reporting approaches***, which limit the ability to consistently quantify emission reductions, manage double-counting risks, and convert climate outcomes into monetizable carbon assets at scale. This gap is particularly evident in landscape-level programmes such as **FOLUR**[12]**,** where restoration outcomes are distributed across thousands of smallholders, and in clean cooking initiatives like **the eQub Clean Cooking Alternative Financing Project (CCAFP)**, where adoption and sustained usage must be credibly tracked across low-income households.

Institutionally, key actors include the **Ministry of Planning and Development**, which is leading national carbon market coordination and MRV system development; the **Environmental Protection Authority**, which provides regulatory oversight and environmental integrity; the **Ministry of Finance**, **Ethiopian Forest Development**, sector ministries, cooperatives, fintech providers, and development partners, including **UNDP Ethiopia**. While these actors are aligned around shared climate goals, they currently lack an **integrated, digital verification layer** capable of connecting project-level data with national systems and international market requirements.

It is within this context that the Kijani MRV solution is proposed. By addressing the digital verification gap without displacing national authority or systems, the platform responds directly to Ethiopia's most pressing climate finance challenge: **how to transform real, on-the-ground mitigation outcomes into trusted, investable climate assets**. Doing so will strengthen carbon market integrity, attract international investment, and channel climate finance to vulnerable communities, while remaining fully aligned with national strategies such as the **Climate Resilient Green Economy**

---

[7] **Taskforce on Scaling Voluntary Carbon Markets (TSVCM) / McKinsey** (2021–2023).
[8] **Ministry of Water and Energy (2023–2024).** *National Clean Cooking Program & National Clean Cooking Roadmap (2026–2035).*
[9] **Government of Ethiopia (2024).** *Carbon Market Regulation.*
[10] **Office of the Prime Minister of Ethiopia (2019–2024).** *Green Legacy Initiative Reports.*
[11] **World Bank Carbon Initiative for Development (Ci-Dev) & World Vision Ethiopia** (2022–2024).
[12] **UNDP Ethiopia (2022).** *FOLUR Project Document (GEF-funded).*

(CRGE) Strategy[13], the **10-Year Development Plan**[14], the **Ethiopia Integrated Sustainable Financing Strategy (2025-2030)** and **Digital Ethiopia 2025**[15].

1.3 Alignment with UNDP Country Office and National Development Priorities

This pilot is strongly aligned with **UNDP Ethiopia's strategic priorities** on climate resilience, innovative finance, and digital transformation, as articulated in UNDP Ethiopia's Country Programme framework[16]. By design, the Kijani MRV solution supports Ethiopia's implementation of its international climate commitments under the Paris Agreement, particularly the enhanced targets in Ethiopia's updated Nationally Determined Contribution (NDC) 2021[17]. The timing of the pilot coincides with Ethiopia's efforts to build an enabling environment for carbon market participation, including the recent 2024 Carbon Market Regulation and associated institutional arrangements for carbon credit approval and trading. In other words, Kijani MRV comes as Ethiopia is putting the policy pieces in place for climate finance, ensuring the solution is highly relevant to current national priorities[18].

Within this context, Kijani MRV is positioned as a **digital verification layer** that strengthens the integrity, transparency, and auditability of project-level MRV outputs while remaining fully compatible with government-led standards and reporting structures. The platform does not create a parallel system; rather, it complements and feeds into Ethiopia's evolving national MRV and registry framework. By generating tamper-proof, well-structured data on emission reductions, Kijani MRV will enhance the credibility of climate projects without bypassing national oversight. This approach aligns with Ethiopia's Climate Resilient Green Economy (CRGE) strategy and Digital Ethiopia 2025 vision, both of which emphasize leveraging technology and innovative finance to achieve climate and development goals.

The pilot intentionally focuses on two high-priority sectors for Ethiopia: **forest-landscape restoration** and **clean cooking energy**. By using two complementary test cases consisting of (1) the FOLUR project and (2) eQub's clean cooking initiative. These were selected because they map directly to Ethiopia's national climate and development priorities: sustainable land management (for emissions reductions and green growth in rural areas) and household energy transition (for health, environment, and women's empowerment co-benefits). Under the FOLUR program, UNDP and partners are working to restore degraded coffee landscapes and promote deforestation-free agriculture in key regions. Achieving national targets in this sector requires accurately measuring carbon sequestration and avoided emissions across dispersed smallholder farms. Strengthening MRV in this context is essential to demonstrate credible performance, reduce MRV-related transaction costs, and unlock results-based climate finance linked to verified land-use outcomes. In parallel, the pilot integrates the **eQub Clean Cooking Alternative Financing Project (CCAFP)** to show how digital MRV can support results-based finance and integrity in household energy interventions. In clean cooking projects, it is not enough to distribute improved stoves; carbon crediting demands proof that stoves are actually being used correctly and consistently over time. Indeed, ensuring adoption and sustained use of clean cookstoves is a known verification challenge in clean cooking markets. Ethiopia's national clean cooking agenda, as outlined in the National Clean Cooking Roadmap (2026–2035), underscores the importance of expanding access to clean cooking solutions and of

---

[13] **Government of Ethiopia (2021).** *Climate Resilient Green Economy (CRGE) Strategy.*
[14] **Government of Ethiopia (2021).** *Ten-Year Development Plan (2021–2030).*
– Source for national development alignment.
[15] **Ministry of Innovation and Technology (2020).** *Digital Ethiopia 2025 Strategy.*
[16] **UNDP Ethiopia.** *Country Programme Document (CPD) for Ethiopia (2021–2025).* United Nations Development Programme, Addis Ababa.
[17] **Government of Ethiopia.** *Updated Nationally Determined Contribution (NDC).* Submission to the UNFCCC, 2021.
[18] **Government of Ethiopia.** *Carbon Market Regulation.* Federal Democratic Republic of Ethiopia, Addis Ababa, 2024.

credibly tracking their usage and impacts over time. By incorporating the eQub pilot, Kijani MRV demonstrates how a digital layer can strengthen evidence of stove usage, improve transparency in emission reduction calculations, and support trustworthy benefit-sharing with low-income households and communities.

Overall, the Kijani MRV pilot supports Ethiopia's long-term direction of combining climate action, digital innovation, and new financing models. It has been explicitly designed to **complement, not replace** national MRV and registry functions by generating verification outputs that can integrate into official systems through agreed protocols and data-sharing arrangements. This ensures that as Ethiopia's government builds its own MRV infrastructure, the lessons and data from Kijani MRV can be absorbed and scaled within national frameworks. The pilot thereby aligns with both UNDP's programmatic focus and the country's development strategies, positioning itself as a catalyst to accelerate Ethiopia's climate agenda in a manner that is locally owned and globally credible.

1.4 Relevance to UNDP Programmes and Resource Mapping

Kijani MRV has been designed as a pilot initiative that can be easily and deliberately embedded within UNDP Ethiopia's existing programme architecture. The design process has attempted to leverage UNDP's portfolios on climate resilience, environmental management, innovative finance, and digital public infrastructure. By anchoring the pilot in the ongoing FOLUR project and the eQub clean cooking initiative, the solution can be tested and refined in real operational environments without creating new parallel structures. This integration ensures that the pilot directly contributes to and learns from active UNDP-supported projects on the ground.

Under FOLUR, UNDP Ethiopia and partners are implementing landscape-level interventions aimed at reducing deforestation, restoring degraded lands, and strengthening sustainable coffee value chains across multiple regions. These interventions are already generating measurable climate mitigation outcomes (e.g. carbon stored in growing forests and soil, or emissions avoided through climate-smart agriculture). However, the ability to attract results-based climate finance or carbon credit income from these achievements is constrained by complex verification requirements, high transaction costs, and fragmented data systems that make it difficult to aggregate and certify the results. By integrating Kijani MRV into FOLUR's operations, UNDP can test how a blockchain-enabled verification platform might reduce MRV "friction" (streamlining data collection and validation), improve traceability of results, and support the future monetization of verified outcomes – all without altering the project's existing design or institutional roles.

In parallel, eQub's Clean Cooking Alternative Financing Project aligns closely with UNDP's priorities on clean energy access, inclusive finance, and digital innovation. eQub's fintech-driven model addresses household energy affordability by digitizing the traditional **equb** (rotating savings and credit) mechanism, allowing low-income families to pay for improved cookstoves in small installments. This approach is breaking financial barriers to clean cooking adoption. However, like many clean cooking programmes globally, linking stove dissemination to verifiable, long-term emission reductions remains a bottleneck for accessing carbon finance. Integrating Kijani MRV into eQub's programme allows UNDP to test how digital verification can strengthen clean cooking MRV by reliably monitoring stove usage and impacts thereby enhancing transparency and ensuring that carbon credit benefits (or other results-based payments) can flow back to the communities using the stoves. It also helps demonstrate an innovative finance model (combining microfinance and carbon incentives) that could be replicated or scaled up through UNDP's global networks.

From a resource mapping perspective, the pilot maximizes the use of **existing resources and partnerships** rather than requiring entirely new funding streams. The FOLUR project provides institutional coordination, baseline data (e.g. on land use and carbon stocks), and field implementation capacity in the target landscapes. The eQub initiative contributes a functioning clean cooking

distribution and finance platform with an established user base of households. The SDG Blockchain Accelerator (through which if this pilot is funded) brings in the technical development support for the digital MRV tool. By piggybacking on these initiatives, Kijani MRV can demonstrate impact cost-effectively. Looking ahead, if the pilot proves successful, its expansion could tap into various climate finance instruments and partners, including for example, UNDP's Biodiversity Finance Initiative (BIOFIN) and other innovative climate finance programs, the Global Environment Facility, bilateral donors (such as NORAD), or even the revenues from carbon markets themselves. In this way, Kijani MRV acts as a cross-cutting enabler that amplifies ongoing UNDP programme results while creating a **replicable digital public good** for Ethiopia's broader climate finance ecosystem.

1.5 Key Stakeholders & Potential Partners

The success of the Kijani MRV pilot depends on a multi-stakeholder partnership model that reflects Ethiopia's climate governance structure and ensures strong community-level engagement. Key stakeholders and potential partners include:

- **Government Ministries and Agencies:** The Government of Ethiopia plays a central role in providing policy oversight, regulatory alignment, and ultimately in adopting and scaling the solution. **The Ministry of Planning and Development (MoPD)** is a crucial counterpart, given its mandate to coordinate national MRV development and carbon market readiness, including ensuring alignment with Article 6 of the Paris Agreement**. The Environmental Protection Authority (EPA)** will provide technical oversight to guarantee that the MRV methodologies and results meet national standards and international carbon crediting criteria**. The Ministry of Water and Energy** is another important partner for the clean cooking component, ensuring that the pilot aligns with Ethiopia's clean cooking policies and the National Clean Cooking Programme framework. Other government bodies, such as the Ministry of Finance and regional environmental bureaus, may be involved to facilitate integration with national systems and budgets.

- **UNDP and Development Partners:** UNDP Ethiopia serves as the **Challenge Owner** and convening agency for this pilot, capitalizing on its role as an implementer of climate initiatives and an honest broker among stakeholders. Through the FOLUR Project Management Unit, UNDP can facilitate access (*upon agreement to take part in this pilot*) to the project sites (e.g. the coffee-growing woredas targeted for restoration), coordinate engagement with regional and local (woreda) authorities, and ensure the pilot complements ongoing land-use and restoration efforts in those areas. UNDP's involvement has the potential to guarantee that the pilot is well-integrated with broader development programs and that knowledge is shared with the government and other UN agencies. The UNDP Country Office can also draw on its global network and experts (for example, the UNDP Climate Promise team or fintech specialists) as needed during implementation. Additionally, other development partners working on climate finance in Ethiopia (such as the World Bank, GIZ, or bilateral donors) are potential collaborators, especially in aligning the pilot with larger climate finance readiness projects.

- **Local Communities and Civil Society:** At the community level, local organizations are indispensable for on-the-ground delivery and ownership. In the FOLUR sites, coffee farmer cooperatives and forest user groups are key implementation partners. They will contribute field-level data (e.g. monitoring tree planting or forest regeneration), help validate the restoration outcomes recorded by the system and participate in any benefit-sharing mechanisms (such as receiving a portion of carbon revenue for community projects). Their involvement ensures that the MRV process is inclusive and that communities understand and trust the system. For the clean cooking component, local women's groups, savings groups, or community health workers could play a role in training users and verifying stove usage in households. Civil society organizations focused on environment, energy, or rural development may also support community outreach and independent monitoring.

- **Private Sector and Technology Partners:** The primary technology/implementation partner on the clean cooking side is **eQub Financial Technologies**, a local fintech company. As the co-solution makers (*upon agreement to partake in this pilot*), eQub provides the digital platform that enables low-income households to acquire improved cookstoves on an affordable payment plan. It will integrate the Kijani MRV tools into its platform to track stove distribution and usage. Local clean cookstove manufacturers and distributors are also important, as they ensure supply of quality stoves and handle last-mile delivery and maintenance. In the forestry sector, private sector involvement may include tech providers for remote sensing or mobile data collection that feed into Kijani MRV. Moreover, mobile network operators and mobile money providers are potential partners to facilitate the flow of carbon finance payments to communities – for instance, using mobile wallets to distribute incentives or revenue shares directly to farmers and households. Engaging these technology and finance providers will help create a robust, user-friendly system and transparent benefit distribution channels.

- **International Carbon Market Institutions:** To ensure the pilot's approach meets global standards, partnerships with international carbon crediting bodies and initiatives are considered. Standards organizations such as **Verra** and the **Gold Standard** can provide methodological guidance, helping align Kijani MRV's data and processes with requirements for issuing verified carbon units. The pilot will draw on criteria from the Integrity Council for the Voluntary Carbon Market (ICVCM) to ensure that any credits generated are of high integrity and eligible in major markets. Collaboration with these bodies not only lends credibility but could also simplify the eventual listing or registration of carbon credits from the pilot projects. Additionally, knowledge exchange with other countries' Article 6 pilots or digital MRV initiatives (through South-South cooperation or UNDP's networks) could be pursued. Together, these stakeholders form a governance and delivery ecosystem that balances national leadership, technical rigor, and community ownership – all of which are critical for demonstrating a scalable and trustworthy digital MRV model.

1.6 Expected Impact

The Kijani MRV pilot is expected to deliver impacts at 3 levels: **systemic**, **community**, and **market**, that extend well beyond the Accelerator period.

System-level impact: The pilot will demonstrate a practical digital MRV layer capable of producing structured, auditable emissions data that can be integrated into Ethiopia's national MRV and carbon market architecture. By generating reliable, tamper-proof records of emission reductions, the solution will improve transparency and trust in climate reporting. It will also help reduce the risk of double-counting emission reductions by linking project data with a unified ledger. Ultimately, a successful pilot will increase confidence among regulators and market participants (buyers, investors) that Ethiopia's carbon credits are backed by high-quality data and oversight. This contributes directly to stronger environmental integrity and governance of carbon markets in Ethiopia.

Programme and community-level impact: Across the two pilot testbeds, Kijani MRV will support measurable climate and development outcomes for local communities. In the FOLUR component, the pilot will target the verification of restoration and sustainable land management activities over an area of approximately 5,000–10,000 hectares of coffee landscapes, involving multiple cooperatives and hundreds of smallholder farmers. By digitally tracking tree planting, forest regrowth, and soil conservation efforts, Kijani MRV will help quantify the carbon sequestration and avoided emissions achieved on these lands. These verifiable emission reductions can then be packaged for future carbon credit issuance, creating a potential revenue stream to further fund community-led conservation. In the eQub clean cooking component, the pilot will monitor the adoption and sustained use of improved cookstoves among roughly 3,000–5,000 households (a scale chosen to match current implementation

capacity). Verified stove usage data, for example, usage frequency and fuel savings captured via the digital app and occasional sensor readings will provide proof of the resulting emission reductions (from less firewood consumption). This not only supports the generation of carbon credits but also correlates with positive co-benefits: reduced household air pollution and health risks, decreased pressure on local forests for woodfuel, and lower cooking fuel expenses for families. In both cases (FOLUR and eQub), the pilot will empower communities with better information (through feedback dashboards or reports) about their contributions to climate mitigation, potentially strengthening local engagement and pride in these efforts.

Market-level impact: By linking robust digital MRV with Ethiopia's climate initiatives, the pilot will help establish a pathway for turning verified emissions outcomes into high-integrity, market-ready carbon assets. This positions Ethiopia to tap into voluntary carbon markets (and future compliance markets via Article 6) more effectively as its national systems mature. In practice, the data and experience from Kijani MRV could inform the design of Ethiopia's national carbon registry and transactional platforms, ensuring they meet the expectations of international buyers and integrity bodies like the ICVCM. Demonstrating successful, transparent credit generation in the forestry and clean cooking sectors could attract greater investment into these areas, both from carbon finance and from climate-aligned development funding. Moreover, by integrating climate action with innovative finance (such as the equb model) and digital technology, the pilot contributes to multiple Sustainable Development Goals, notably **SDG 13 (Climate Action)**, **SDG 15 (Life on Land)**, and **SDG 7 (Affordable and Clean Energy)**. The knowledge generated will be documented and shared, so that it can be replicated in other regions of Ethiopia and inform UNDP's global work on digital climate solutions. Ultimately, the expected impact is a proof-of-concept that shows how a country like Ethiopia can leverage blockchain and digital tools to enhance the credibility and financing of its climate actions, driving both national development and global emission reduction efforts.

2. User Mapping & Testing Strategy

2.1 Primary User Personas

Tesfaye, 42, is the manager of a coffee cooperative in Bensa, Sidama Region. He oversees the cooperative's operations, coordinating reforestation activities across dozens of smallholder farmers and ensuring compliance with sustainability standards. Tesfaye's main challenge is maintaining accurate, verifiable data on tree planting and deforestation-free practices while meeting reporting requirements for carbon credit eligibility. He needs a user-friendly digital platform that allows him to track progress across multiple farmer groups, upload evidence (e.g. geotagged photos from field agents), monitor verification status in real time, and confirm when carbon revenues are disbursed to the cooperative. For Tesfaye, the solution must integrate seamlessly with mobile money systems to facilitate transparent payments and provide dashboards that help him communicate results to cooperative members and coffee buyers.

Aster, 34, is a mother of four and the leader of a women's savings group in a rural community of Jimma Zone, Oromia Region. Her family relies on a traditional three-stone fire for cooking, which consumes substantial firewood and exposes her and her children to smoke. Through the new clean cooking initiative, Aster's savings group is financing improved cookstoves for its members. Aster's challenges include ensuring that all members consistently use their new stoves and tracking these usage patterns to qualify for carbon credit benefits (which can reduce the effective cost of the stoves for her community). She needs a straightforward mechanism, such as a basic mobile app or SMS-based system to regularly

report stove use and maintenance status. Aster also wants to see when her group's clean cooking efforts are verified and carbon incentives are earned, so she can plan group repayments and demonstrate the benefits (like fuel savings and improved health) to her community. For her, the solution must be intuitive in the local language and functional even in low-connectivity areas, ensuring that even remote households can be included.

2.2 User Story

"**As Tesfaye**, a cooperative manager, **I want to** track reforestation progress across multiple farmer groups, see verification status in real time, and know when carbon revenues are disbursed to my cooperative **so that** I can manage finances transparently, maintain accountability, and effectively report results to members and buyers".

"**As Aster**, a clean cooking program participant, **I want to** easily confirm my household's stove usage (for example, by submitting periodic usage reports or sensor data) and be notified when our usage is verified **so that** my women's group can qualify for carbon credits that lower our stove payments and I can reassure my family and community that our efforts are recognized and rewarded".

2.3 Testing Strategy

The testing strategy for the Kijani MRV platform is structured to validate its usability, technical robustness, and policy alignment under real-world conditions over a 12-month pilot period. This pilot is proposed to take place in two distinct geographies: Bensa (Sidama) and a selected woreda in Jimma Zone (Oromia) representing different climate, livelihood, and infrastructure contexts. Bensa offers a forest-based coffee agroforestry landscape suitable for the land-use application under FOLUR, while the Jimma pilot site (tentatively identified and subject to final confirmation) will represent the household energy transition use case, supporting eQUB's clean cooking initiative as it scales beyond Addis Ababa in 2026.

The sites are selected to reflect diverse agro-ecological zones and end-user profiles, enabling robust testing of Kijani MRV's adaptability to both communal agroforestry and household-level clean energy deployment. The pilot will be deployed in a phased, iterative manner to enable technical feedback, institutional learning, and user capacity development.

Phase 1: System Onboarding & Stakeholder Readiness (Months 1–2)

This phase focuses on preparation and groundwork. Activities include:

- Partner validation of pilot locations and site-specific needs assessments
- Mapping of user groups, including cooperative leaders, enumerators, and savings group coordinators
- Training workshops and onboarding sessions on MRV data collection workflows, mobile use, and platform access
- Establishing feedback loops and coordination protocols with local stakeholders (e.g. cooperative boards, field officers, community agents)

Phase 2: Usability Testing (Months 3–4)

Low-fidelity wireframes and clickable prototypes of the mobile interface and dashboard tools will be tested with core user groups. Key objectives:

- Validate navigation flow, language localization, and mobile accessibility
- Assess comprehension of MRV workflows, from field data entry to verification approval
- Identify pain points in user onboarding and offline data entry for areas with limited connectivity
- Feedback is captured through structured surveys and in-app analytics

Phase 3: Field-Based User Acceptance Testing (UAT) (Months 5–7)

A functional mobile application integrated with satellite analytics and geospatial tagging will be deployed for real-time data collection. Project enumerators and cooperative focal points will:

- Capture plot-level geocoordinates (for restoration sites) and household data (for clean cooking users)
- Upload timestamped photos (e.g., tree planting, stove installation, usage metrics)
- Submit verifiable MRV records with integrity metadata (anti-tampering hashes, location data)
- Test mobile-to-registry synchronization under intermittent connectivity

This phase will test interoperability with mobile payment systems for conditional payouts and trigger alerts to the dashboard based on flagged data or non-compliance.

Phase 4: Controlled Registry Simulation (Months 8–9)

A full lifecycle simulation will be conducted on a blockchain testnet to mimic issuance, verification, and retirement of carbon credits:

- Project registration on the testnet
- Submission and review of field MRV packages
- Issuance of dummy carbon credits
- Simulation of credit transfers and final retirement (mimicking buyer-side purchase)

This will validate the end-to-end traceability of carbon transactions and confirm the platform's compliance with audit and transparency requirements.

Phase 5: Buyer-Side Provenance Review (Months 10–11)

Select observers and prospective buyers or auditors will receive read-only access to the testnet buyer portal:

- Review of verification lineage, issuance logic, and proof-of-retirement
- Validation of the transparency, auditability, and completeness of MRV records
- Feedback captured through buyer-side scoring rubrics and integrity checks

Phase 6: Final Review and Pilot Assessment (Month 12)

The final month will be dedicated to:

- Synthesizing user feedback and buyer input
- Benchmarking technical performance (uptime, data lag, issue resolution)
- Documenting learnings and preparing for post-pilot refinement

Throughout the year-long pilot, weekly syncs and monthly coordination meetings will ensure iterative learning. All feedback will be logged and incorporated in agile development sprints.

Note: Final pilot locations remain subject to confirmation with implementing partners (FOLUR and eQUB) based on their operational readiness. The current selections—Bensa and Jimma—are proposed based on programmatic relevance, anticipated user volume, and agro-ecological diversity.

2.4 Feedback Loop Design

Kijani MRV incorporates a structured feedback loop to drive continuous improvement and maintain alignment with stakeholder needs and policies during the pilot. Feedback mechanisms are embedded at multiple levels of development and testing:

1. **Field Sprint Feedback:** After each field-testing sprint (e.g. post-UAT deployment in a community), structured feedback is collected from participants through short user surveys (capturing quantitative usability metrics) and semi-structured interviews with key users like project officers, enumerators, cooperative managers, and savings group leaders. This frontline feedback focuses on practical issues such as usability hurdles, data capture challenges (for instance, difficulties in uploading photos or stove usage logs), clarity of workflows, and any operational constraints observed on the ground.

2. **Weekly Project Management Unit (PMU) Sync:** A weekly internal coordination meeting (the PMU sync) reviews all incoming feedback and bug reports. The team categorizes feedback items by severity (critical, high, medium, low) and logs them in a central tracker. Actionable issues are then prioritized into mini sprints for resolution. For example, if field testers report that the app's interface is confusing in the Afan Oromo language setting, or that offline data upload failed in a remote village, these would be flagged as high priority to fix before the next release. This rapid response cycle ensures that the most pressing technical issues and user experience improvements are addressed on an ongoing basis.

3. **Fortnightly Policy & Governance Clinics:** Every two weeks, the team conducts structured sessions with relevant government and institutional stakeholders to validate the pilot's alignment with national MRV standards and policy expectations. Key proposed participants include the **Environmental Protection Authority (EPA),** the **Ministry of Planning and Development (MoPD), Ethiopian Forest Development (EFD)**, **and the Ministry of Water and Energy (MoWE)** , the latter to ensure the clean cooking component remains in sync with Ethiopia's national energy and clean cooking policies. In these clinics, the pilot team presents recent developments (e.g. how field data is being recorded and secured) and stakeholders examine aspects like data governance, registry interoperability with national systems, and Article 6 (Paris Agreement) compliance. Any policy concerns or suggestions (for instance, data privacy safeguards or approval workflows required by regulators) are fed back into the system design.

4. **Change Management & Documentation:** All approved changes to the system or processes are implemented through controlled mini-sprints and documented in release notes. Version updates and their implications are communicated to stakeholders regularly during PMU syncs and policy clinics. This transparent change log and communication flow ensures everyone from developers to government partners shares a common understanding of how the system is evolving over the pilot period.

Through this multi-tiered feedback loop, the Kijani MRV pilot maintains a disciplined yet adaptive development process. It ensures that on-the-ground learnings and stakeholder inputs continuously inform the platform's refinement. Ultimately, this approach keeps the solution aligned with real-world

user needs and regulatory requirements, while building confidence among partners that the system can scale beyond the pilot with integrity and success.

3. Solution Overview

3.1 Solution Summary

Kijani MRV is a blockchain-backed **Measurement, Reporting, and Verification (MRV)** platform designed to support Ethiopia's emerging carbon market by creating a transparent national carbon credit registry and marketplace. The one-year pilot will demonstrate the solution across two proposed projects **(a)** a clean cooking project led by eQUB (distributing fuel-efficient cookstoves), and **(b)** a forestry and coffee landscape restoration project under the FOLUR program which are both aligned with Ethiopia's climate priorities.

The platform addresses critical gaps: Ethiopia faces a massive **$316 billion climate finance gap** to meet its 2030 NDC targets, even as global demand for high-quality carbon credits is projected to reach $250 billion annually by 2030. Without transparent and trusted systems for MRV and credit issuance, the country risks being left behind in this fast-growing market. Kijani MRV aims to turn this risk into opportunity by **tokenizing verified emission reductions on Cardano's blockchain**, ensuring each ton of CO_2 is tracked and monetized with integrity. By **improving MRV capacity and data transparency**, the solution supports Ethiopia's new carbon market strategy which calls for robust MRV systems and governance reform. It directly contributes to SDG targets: **SDG 13 (Climate Action)** and **SDG 7 (Clean Energy)** through reduced emissions from clean cookstoves, and **SDG 15 (Life on Land)** via forest restoration. Co-benefits include health improvements and gender impacts (clean cooking reduces indoor air pollution and eases burdens on women), as well as potential peacebuilding dividends by engaging ex-combatants and youth in green jobs (e.g. as forest rangers or stove distributors). In summary, **Kijani MRV** will pilot a secure, inclusive carbon finance ecosystem that channels global climate funding to local communities, while building Ethiopia's reputation as a supplier of high-integrity carbon credits in Africa.

3.2 Core Features and Functionalities

Kijani MRV's design brings together digital monitoring tools and blockchain smart contracts to ensure end-to-end transparency and efficiency in the carbon credit lifecycle:

1. **Digital MRV Data Collection:** The platform integrates **IoT sensors, satellite remote sensing, and mobile data apps** for real-time monitoring. For example, cookstove usage is tracked via sensor or survey data, and forest cover/regrowth is monitored through satellite imagery. An AI-powered analytics engine processes this data to quantify emissions reduced or CO_2 sequestered, dramatically cutting the time and cost of verification. This **automated MRV** approach (inspired by solutions like EcoNetix's digital MRV) utilizes software + hardware + ML models to simplify and strengthen verification processes, reducing reliance on expensive third-party auditors.

2. **Blockchain Credit Registry:** All verified emissions data feeds into a **Cardano blockchain registry** that issues a corresponding carbon credit token for each certified tone of CO_2. **Smart contracts** on Cardano enforce that credits are **issued, traded, and retired** in a transparent, tamper-proof manner, preventing double-counting or fraud. Each credit token carries metadata

on its project origin, vintage, and SDG impact, ensuring traceability for buyers. The registry acts as a single source of truth accessible to regulators and participants, aligning with Ethiopia's push for transparent approval and accounting in carbon trading.

3. **Carbon Marketplace & Exchange (Future Module):** Kijani MRV, in its later phase includes a user-facing marketplace module where carbon credits can be listed and transacted. This marketplace will connect project developers (sellers) with buyers/investors seeking high-integrity Ethiopian credits. Transactions are settled in near real-time via Cardano, with low fees and secure digital wallets. **Trust and price discovery** are enhanced by on-chain records (each token trade is publicly verifiable), attracting buyers who demand clear provenance of credits. This feature supports Ethiopia's goal to mobilize climate finance via Article 6 and voluntary markets.

4. **Inclusive Finance Mechanisms:** Uniquely, the solution is built to **directly reward local communities** generating the climate benefits. Smart contracts will automatically allocate a share of credit sale revenues to stakeholders like rural households, women's cooperatives, or groups participating in the projects. For instance, when an issued cookstove credit is sold, a programmed percentage of the payment could be disbursed to the stove user or community fund. This kind of **"payment-for-results"** mechanism, enabled by blockchain, ensures fair benefit-sharing and incentivizes grassroots adoption of climate-friendly practices.

5. **User Dashboard and Alerts:** The platform provides a web and mobile dashboard for different user roles – project implementers can upload field data and see their credit generation in real time; verifiers can review data flagged by AI for quality assurance; government regulators can view the registry and approve issuances; and buyers can browse available credits with confidence in their quality. Automated alerts/notifications are built in (e.g. if a sensor shows a stove is inactive, or if a credit is nearing expiration) to enable proactive management. Role-based access and a multilingual interface will ensure accessibility even in low-connectivity settings (e.g. SMS or offline data capture options, important for rural Ethiopia).

6. **Integration & Compliance:** The system is designed to **integrate with national climate registries and standards**. It follows open API standards so that, as Ethiopia develops its national carbon registry law and digital infrastructure (per the 2024 Carbon Market Regulation and digital directives), Kijani MRV can plug in seamlessly. All methodologies will align with recognized carbon standards (e.g. Gold Standard or Verra), ensuring the credits are **high-integrity and internationally marketable**. The platform's security architecture (leveraging Cardano's robust proof-of-stake consensus) and data encryption will comply with data protection and financial regulations. This compliance-by-design approach addresses any legal or policy barriers early (see Section 5.11), giving regulators confidence in the pilot.

3.3 Tech Stack and Tools

Kijani MRV is built on a secure, interoperable, and future-proof technology stack. The design supports high-integrity carbon market operations, works in low-bandwidth rural environments, and can scale across different geographies. Key components include:

- **Blockchain & Ledger Layer:**
    - **Primary Chain – Cardano:** The solution runs on the Cardano blockchain (using a testnet during the pilot phase with a planned migration path to mainnet[1]). Cardano's proof-of-stake infrastructure provides security and scalability for the carbon registry.

- **Anchoring Strategy:** The platform periodically anchors the state of the registry and MRV data hashes to a secondary blockchain. This approach enhances resilience, cross-platform interoperability, and long-term auditability of records.

- **Smart Contracts in Aiken:** All carbon credit issuance, transfer, and retirement logic is implemented via Cardano smart contracts written in Aiken[2]. These contracts enforce business rules (e.g. preventing double issuance or unauthorized transfers) and uphold the integrity of the registry.

- **Decentralized Storage:**

  - **IPFS/Filecoin:** MRV evidence (satellite analysis outputs, geo-tagged field photos, field measurements, verifier reports, etc.) is stored on decentralized storage networks like IPFS and Filecoin. This off-chain storage ensures large files are accessible and shareable without bloating the blockchain.

  - **On-Chain References:** Instead of storing bulk data on-chain, only cryptographic references (e.g. Merkle root hashes of the evidence files) are recorded on the blockchain. This makes the records tamper-evident – any change to the off-chain files would be detectable via the hash – while keeping sensitive data private and the on-chain footprint small.

- **Identity & Access Management:**

  - **Atala PRISM Decentralized IDs:** The platform uses Atala PRISM for decentralized identity management[3]. All key actors (project developers, verifiers, the Program Management Unit, etc.) receive decentralized identifiers (DIDs) and verifiable credentials attesting to their role and permissions.

  - **Credential-Based Access Control:** System permissions and workflows are enforced based on these DID-linked credentials. In practice, only an entity with the appropriate verifiable credential (e.g. an accredited verifier's DID) can perform certain actions on the platform. This mechanism ensures that data submission, verification approvals, and credit issuances are only carried out by authorized, accountable parties – with each action transparently tied to a credential.

- **AI, Analytics & Geospatial Intelligence:**

  - **Machine Learning Models:** Kijani MRV incorporates AI models to analyze environmental data. For example, the system can automatically detect changes in forest canopy or land cover to verify reforestation and agroforestry project outcomes. Similarly, specialized models monitor coffee farming areas to identify rejuvenation of coffee gardens and changes in productivity, supporting climate-smart agriculture metrics.

  - **Geospatial Analytics:** The platform provides spatial analysis tools such as map overlays, time-series comparisons of land use, and boundary delineation analysis. These analytics are delivered via interactive dashboards, helping stakeholders observe trends like forest regrowth or land-use changes over time in each project area.

  - **Visualization Dashboards:** User-friendly geospatial dashboards allow regulators, verifiers, and buyers to visually inspect project locations and their surroundings. They can review land-use changes, verification site visits, and historical performance at a glance. This visual layer makes complex MRV data easier to interpret and builds trust through transparency.

- **Backend & Application Layer:**
    - **Core Languages:** The solution leverages a mix of programming languages, each chosen for specific purposes. **TypeScript/JavaScript** power the core backend services, APIs, and dynamic frontend behavior. **Python** is used for the AI/ML pipelines, satellite image processing tasks, and data analytics. **Aiken** is used for on-chain smart contracts (as noted above). **PL/pgSQL** (PostgreSQL's procedural SQL) is employed for database routines and efficient spatial queries. **Shell scripting** is utilized to automate DevOps tasks, deployments, and continuous integration/ continuous deployment (CI/CD) workflows.
    - **Architecture:** The backend follows a modular, API-driven design. It interfaces with the blockchain layer, the identity system, and the analytics engine, abstracting away complexity for the end user. A web-based frontend and a possible mobile interface (for field data capture) make the system accessible to users in the field and in offices alike.
- **External APIs & Integrations:**
    - **Blockfrost API:** The platform uses Blockfrost[4], a Cardano API service, to interact with the blockchain. This API handles transaction submissions, queries the state (e.g. UTxOs and smart contract data), and monitors events without requiring Kijani to run its own full Cardano node.
    - **Sentinel Satellite Data:** Integration with Sentinel Earth observation data provides timely satellite imagery and environmental datasets for MRV analysis. For instance, the system can pull Sentinel-2 imagery (multispectral satellite images) to assess vegetation health or land changes in project areas. This built-in integration ensures the MRV engine has up-to-date, reliable remote sensing inputs for verification.
    - **Mapbox:** Mapbox services are used for mapping and geospatial visualization in the user dashboards. This allows interactive maps where users can select GPS coordinates, draw project boundary polygons, and visualize geospatial data. Mapping integration enhances user experience by making spatial data exploration intuitive (e.g. panning/zooming on project sites, viewing heatmaps of activity).

All these layers work together to ensure Kijani MRV remains **secure, transparent, and scalable**. The stack is compliant with international MRV and carbon-market standards, yet is designed to be practical for local users. By using familiar web and mobile tools on the front end and robust blockchain/AI services in the back end, the platform makes advanced technology accessible to project developers and regulators on the ground.

3.4 Blockchain Functionality

Kijani MRV leverages blockchain as a **trust and integrity layer** rather than a user-facing feature. In other words, the blockchain is working behind the scenes to provide security and transparency, without adding complexity for the end user. The blockchain components ensure end-to-end auditability and alignment with international carbon market standards, all while keeping the user experience simple and approachable.

Key blockchain functionalities include:

- **Smart Contracts for Issuance & Retirement:** Cardano smart contracts govern the issuance of new carbon credit lots and the retirement of used credits. After a project's emission reductions are successfully verified (via the MRV process), a smart contract triggers the issuance of the corresponding digital carbon credits. When credits are to be retired (e.g. a buyer uses the credits to offset emissions), the retirement function is executed on-chain. Once a credit lot is retired, it becomes permanently non-transferable – ensuring those credits cannot be double-counted or resold. This contract-enforced rule preserves environmental integrity by guaranteeing that retired credits represent finalized climate impact.

- **Tokenization of Credit Lots:** Each verified emission reduction outcome is represented on-chain as a **tokenized credit lot**. These on-chain tokens carry metadata linking them to a specific project, a vintage (time period), and the applicable methodology or standard. The tokens can be transacted between authorized parties (e.g. project developers transferring to buyers) seamlessly on the blockchain, up until the point of retirement. Tokenization provides transparency (every token's origin and history is traceable) and efficiency in how credits are issued, traded, and managed.

- **On-Chain Hashing of MRV Dossiers:** For every batch of credits issued, there is a comprehensive MRV dossier (including field measurements, satellite analysis results, verifier reports, etc.). Kijani stores these dossiers off-chain (in the secure evidence repository), but a **cryptographic hash** of each dossier is recorded on-chain. This hash serves as a fingerprint of the documentation. It guarantees that any tampering with the evidence would be detectable if someone altered a report later, the hash would no longer match the on-chain record. Thus, the blockchain provides an immutable audit trail for the MRV evidence, ensuring long-term verifiability **without exposing sensitive data** publicly.

- **DID-Based Authorization:** Interactions with the smart contracts (issuance approvals, retirement executions, etc.) are gated by decentralized identities. Issuers, verifiers, and regulators use their Atala PRISM DIDs to authenticate and authorize actions on-chain. Only entities holding the proper credentials can invoke the smart contract functions to issue or retire credits. Every on-chain action is linked to a DID, meaning there's an auditable record of **who** (which credential) performed each action. This ties into the platform's role-based access control, reinforcing that only accredited actors can generate or modify critical records.

- **Public Proof-of-Retirement Pages:** To enhance transparency for buyers and external stakeholders, the system generates a public-facing proof-of-retirement page whenever credits are retired. This page is essentially a human-readable certificate or record of the retired carbon credits, which can be accessed through a web link. It contains key details like the project name, unique identifiers of the credit lot, the quantity of CO_2 offset, the date of retirement, and references (hashes or IDs) to the verification evidence. Anyone, including buyers or auditors can independently verify a credit's legitimacy and retirement status **without needing a blockchain wallet or technical knowledge**. This feature abstracts the on-chain data into a user-friendly format, building trust with end users by providing easily accessible proof of the climate impact.

Overall, the blockchain functionality in Kijani MRV underpins **trust, traceability, and compliance**. By using Cardano's secure ledger and smart contracts, the platform ensures that every carbon credit issued is backed by verifiable data and that once used, credits cannot be fraudulently reused. At the same time, through DIDs and thoughtful design, these guarantees are achieved without making the system hard to use – participants interact through simple apps and dashboards, while the complex blockchain processes run transparently in the background.

4. Weekly Workplan

The following is a detailed workplan for the 12-month pilot, broken down by week, with two implementation approaches considered (Option A and Option B). Each option sequences the same activities differently to account for uncertainties in pilot site confirmation and resource allocation. The workplan aligns with the deliverables in Section 5 and ensures all stakeholders are engaged throughout.

4.1 Option A – Sequential Pilots (Staged Deployment):

In Option A, we prioritize one pilot site at a time in a staged approach. The clean cooking (eQUB) pilot is launched first (after initial platform development), followed by the forestry/coffee (FOLUR) pilot mid-year. This allows the team to focus efforts and incorporate lessons from the first deployment into the second.

- **Weeks 1–4:** Project kickoff and planning. Formalize partnerships (MoUs with eQUB and FOLUR stakeholders), appoint local project leads for each pilot. Conduct stakeholder workshops to refine requirements and success metrics. *Deliverables:* Stakeholder Engagement Plan and detailed Requirements Document completed by Week 4 (Sections 5.1–5.2).

- **Weeks 5–8: Platform design and prototype development.** Set up cloud infrastructure and blockchain environment (Cardano testnet). Develop core smart contracts for credit issuance and basic front-end interfaces. Begin configuring sensor integrations and satellite data pipelines. Hold design review with UNDP and technical partners (EMURGO) at end of Week 8. *Deliverables:* Solution Architecture & Data Flow Diagram (Deliverable 5.2) by Week 8.

- **Weeks 9–12: MVP completion and internal testing.** Finish MVP (minimum viable product) development: user login, data upload, token minting logic on testnet. Conduct unit tests and fix bugs. Simulate MRV process with dummy data from both use cases to validate end-to-end flow. Prepare training materials for field staff. *Deliverables:* MVP Demonstration (internal) and Test Report by Week 12. Pilot sites confirmed for deployment.

- **Weeks 13–16: Deploy Pilot 1 – Clean Cooking.** Roll out the platform to eQUB's clean cookstove project. Install IoT stove sensors (if used) or train community enumerators on the mobile app for usage data collection. Import baseline data (e.g. families, stoves distributed) into the system. Commence data collection in the field. Monitor data uploads and system performance closely; provide on-ground support to users. *Deliverables:* Pilot Launch Report for Clean Cooking (Deliverable 5.5) by Week 16.

- **Weeks 17–20: Pilot 1 operational and first credit issuance.** Continue running the cookstove pilot; by Week 18, enough usage data is collected to calculate emissions reductions. The MRV engine processes data and generates the first batch of carbon credits (tokens) for reduced firewood use. Verifiers review and approve issuance via the platform. Host a demo day in Week 20 with stakeholders: showcase the live dashboard, and if possible, execute a token transaction (test sale or retirement) to prove the concept. *Milestone:* First carbon credits tokenized from Pilot 1 by Week 20.

- **Weeks 21–24: Evaluation & iteration for Pilot 1; Prep for Pilot 2.** Gather feedback from Pilot 1 participants (users, verifiers, eQUB staff). Assess any technical or adoption issues (e.g. difficulties with the app, data accuracy). Implement improvements: e.g. refine the AI model if needed, enhance user training. In parallel, finalize the implementation plan for Pilot 2 (FOLUR): confirm target sites (forest communities), and ensure required data (satellite

baseline, etc.) is prepared. *Deliverables:* Interim Impact Assessment of Pilot 1 (part of Deliverable 5.8) and Revised Deployment Plan for Pilot 2 by Week 24.

- **Weeks 25–32: Deploy Pilot 2 – Forestry/Coffee Restoration.** Launch the platform in the FOLUR project areas. Train forest rangers or community scouts to use the app for reporting (e.g. tree planting, patrolling) and integrate with any existing FOLUR monitoring systems. Use satellite monitoring to set a baseline for forest cover. By this stage, the platform (with improvements from Pilot 1) is robust. Monitor Pilot 2 closely through its initial 2 months. *Deliverables:* Pilot Launch Report for Forestry (Deliverable 5.5 updated) by Week 28; mid-pilot technical review by Week 32.

- **Weeks 33–36: Pilot 2 operational and credit generation.** By around Week 33–34, initial data on tree survival, growth, or avoided deforestation is analyzed. The platform issues the first batch of forest carbon credits (e.g. for hectares restored or tCO_2 sequestered) using the smart contracts. Coordinate with government bodies (e.g. Ministry of Environment or Planning) to observe the issuance process, reinforcing regulatory buy-in. Meanwhile, Pilot 1 (cookstoves) continues to generate data – possibly a second credit issuance cycle occurs in parallel. Ensure the marketplace is ready to list these credits (even if just demo sales to UNDP or a partner buyer).

- **Weeks 37–44: Combined Pilot monitoring and stakeholder engagement.** Both pilots are now running in parallel. The team conducts ongoing monitoring, verification, and support. Host workshops with communities to explain the link between their actions and earned credits/income (strengthening local buy-in). Begin engaging potential credit off-takers: e.g. carbon market intermediaries or corporate CSR programs, to introduce the forthcoming credits. Any issues in data flow or smart contract performance are addressed in real-time. *Deliverables:* Draft **Impact Measurement Report** (Deliverable 5.8) by Week 44, capturing data from both pilots to date. Also, conduct a **Mid-point Evaluation** meeting with all stakeholders in Week 40 to assess progress against KPIs and adjust if necessary (Deliverable 5.12, risk review).

- **Weeks 45–48: Project wrap-up and knowledge transfer.** As the one-year mark approaches, plan for pilot conclusion. Ensure all data is backed up and shared with stakeholders. Organize capacity-building sessions to enable local partners to continue using or maintaining the system. Develop the **Sustainability Roadmap** (Deliverable 5.9) outlining options for scaling beyond the pilot (e.g. integration into national registry or expansion to new projects), and a **handover plan** if the government will take on the platform. Finalize all documentation (technical manuals, user guides, compliance reports).

- **Weeks 49–52: Final evaluation and reporting.** Conduct a summative evaluation against initial objectives: How many credits were generated? How much income went to communities? What were the SDG impacts (e.g. reduction in wood usage, hectares restored)? Derive lessons learned and best practices. Host a **Pilot Demo Day/Closing Workshop** (Week 50) to present results to the SDG Accelerator, government officials, and potential investors. Collect stakeholder testimonies (e.g. a woman who earned income from stove credits, or a community benefiting from restoration). In the final weeks, focus on dissemination: prepare policy briefs and a slide deck for outreach. *Deliverables:* **Final Pilot Report** and **Impact Dashboard** delivered by Week 52, along with the completed **Deliverables Completion Table** (Section 5.14).

4.2 Option B – Parallel Pilots (Accelerated Deployment):

Option B assumes both pilot sites are confirmed early, allowing an accelerated, parallel implementation to maximize learning within one year. The platform is developed and rolled out in

both contexts with overlapping timelines, requiring a larger team to handle parallel streams. This option can demonstrate scalability but carries higher coordination risk.

- **Weeks 1–4:** Initialization similar to Option A: project kickoff, stakeholder alignment, and requirements gathering. However, in Option B we establish two field teams from the start – one focused on the cookstove pilot, one on the forestry pilot – each with designated coordinators. Dual-track planning workshops are held to capture use-case-specific needs. (e.g. cookstove team refines requirements for household data and payment distribution, while forestry team focuses on satellite MRV integration). *Deliverables:* Complete stakeholder mapping and needs assessment for both pilots by Week 4.

- **Weeks 5–8: Concurrent development tracks.** Core platform development (backend, blockchain contracts, database) is done centrally, but in parallel each pilot team begins preparing their domain: the cookstove team procures sensors and configures the mobile app for households, while the forestry team sets up satellite data feeds and trains on GIS tools. Bi-weekly sync meetings ensure the tech development aligns with both use cases. By Week 8, a unified MVP that supports both cookstove and forestry data structures is ready for testing.

- **Weeks 9–12: Pilot deployments (Phase 1).** In Option B, we initiate both pilots nearly simultaneously once the MVP is tested. Week 9–10: deploy to cookstove communities (perhaps in one region) *and* initiate forestry monitoring in a selected kebele (community) in the FOLUR area. This phased rollout (a few communities first) is to manage load. Field staff are on the ground in both sites helping users. By Week 12, both pilots have ingested initial data (e.g. a week of stove usage data and initial forest patrol logs). The tech team provides remote support to both infield teams.

- **Weeks 13–20: Full pilot operations in parallel.** Both pilots ramp up to full scale (all target households enrolled, all target forest areas under monitoring). The platform handles data from both streams concurrently. This period is intensive: the team splits attention, with a support engineer and community liaison dedicated to each pilot to troubleshoot issues. Despite the split, cross-pilot learning is encouraged (weekly internal stand-ups to share insights). By around Week 18–20, each pilot should achieve its first credit issuance milestone: cookstoves credits minted for initial usage period, and forestry credits minted for initial quarter of forest protection or tree growth. Celebratory events are held in each community to mark these milestones, emphasizing the inclusive finance aspect (households or community groups see the first signs of potential revenue).

- **Weeks 21–28: Monitoring, optimization, mid-point evaluation.** With both pilots fully operational, the focus shifts to optimization and interim evaluation. Data from Weeks 13–20 is analyzed for quality and impact. If one pilot is facing challenges (say sensor connectivity issues or satellite data anomalies), additional resources are temporarily reallocated to resolve them (the advantage of Option B is we can compare performance – e.g., if the cookstove MRV is yielding more reliable data than forest MRV, we investigate and bolster the latter). Conduct a **mid-point evaluation workshop in Week 24** with both pilot stakeholders present, fostering exchange (community members from cookstove pilot meet those from forest pilot, sharing experiences). Update risk logs and mitigation plans (Deliverable 5.12) based on real data. Implement any mid-course corrections (software patches, training refreshers, etc.) by Week 28.

- **Weeks 29–40: Continued operation & integration efforts.** Both pilots continue to generate credits on schedule (likely a second issuance cycle by around Week 36 for each). During this phase, attention is given to external integration: engaging with the government to integrate pilot data into the national MRV or registry initiatives. For instance, by Week 32 we might initiate discussions for the **National Carbon Registry Sandbox** – feeding our token data into the government's sandbox environment (aligned with the National Bank's fintech sandbox

timeline in 2025). Also, approach carbon market platforms or corporate buyers to secure offtake agreements for a portion of the credits (to demonstrate actual market transactions by end of pilot). By Week 40, we aim to have letters of intent or a sample sale executed for each pilot's credits (subject to regulatory approval). Meanwhile, user engagement is maintained: periodic community meetings to ensure sustained participation and to avoid data fatigue.

- **Weeks 41–48: Scale-out planning and preliminary scale activities.** As the pilots near completion, Option B's parallel success positions us to explore scaling earlier. In this window, we identify opportunities to **extend the solution to new sites or stakeholders even before the pilot formally ends**. For example, initial talks with another clean cooking initiative or a REDD+ forestry project could be initiated, using our pilot results as proof-of-concept. The tech team ensures the platform is documented and modular to support adding new projects. Weeks 45–48 are spent drafting the Sustainability & Scale-up Roadmap (Deliverable 5.9 and 5.13), with concrete steps for integration into national systems and expansion to other regions. Both pilot streams continue until at least week 48 to collect full year data (for robust impact analysis).

- **Weeks 49–52: Unified final evaluation and closure.** Conduct a combined evaluation covering both pilots. Because Option B put more strain on resources, the evaluation will look at comparative outcomes (e.g. carbon credits generated, cost per credit, community satisfaction) between the two pilots – generating insights on where the approach is most effective or needs adjustment. Final deliverables (reports, frameworks) are compiled as in Option A, but with dual-site data. The closing workshop (Week 50) will present both pilot stories side by side, potentially with representatives from each community sharing their perspectives. Wrap up all outstanding tasks, ensure each community and stakeholder has a transition plan for after the pilot (e.g. how they can continue monitoring or benefit from any future credit sales). All final documents and the deliverables completion table are completed by Week 52, marking the end of the pilot year.

5. Final Deliverables

5.1 Solution Demo (PoC)

A functional or visual prototype demonstrating the end-to-end workflow of Kijani MRV, from project registration through verification, issuance, and retirement. The demo is presented virtually during scheduled coordination meetings with the CO team and UNDP/Partner focal points.

5.2 Testing & Feedback Documentation

A summary of all testing activities, including usability testing and workflow simulations. Key feedback captured includes:

- The need to abstract wallet interactions away from users and handle blockchain operations in the backend
- Requests for UX simplification and clearer role-based dashboards
- Observations on verifier workflows and approval clarity

Comprehensive documentation is provided for all testing activities conducted during the accelerator. This includes records of user acceptance tests, usability sessions, and simulated end-to-end workflow trials. The report details feedback collected from diverse stakeholders and test users. Key insights from the testing phase included:

- **Blockchain UX:** The need to abstract blockchain wallet interactions away from end-users, instead handling Cardano transactions on the backend to streamline the user experience. This would shield users from technical complexities (keys, fees) while preserving transparency.

- **User Interface & Roles:** Requests to simplify the user interface and provide clearer, role-specific dashboards for different user groups (e.g. project developers, verifiers, regulators). Testers found that role-based views would help each stakeholder focus on their pertinent tasks without distraction.

- **Verification Workflow:** Observations that the verifier workflow needed more clarity in how approvals are submitted, reviewed, and finalized. Both project implementers and third-party verifiers sought a more guided process to reduce ambiguity in the MRV verification and credit issuance steps.

All feedback is captured with timestamps and contributor details, forming the basis for subsequent solution refinements.

5.3 Feedback Integration Summary

A documented account of changes will be made in response to testing and stakeholder feedback, including:

- Backend-managed wallet and transaction handling
- Improved navigation and user flows
- Clarified verification and review processes

A structured summary describes how the pilot team addressed the above feedback through iterative improvements. This “change log” of enhancements demonstrates a responsive development approach aligned with user and stakeholder needs. Notable changes made in response to testing and stakeholder input include:

- **Backend Transaction Management:** Implementing backend-managed wallets and transaction processing to fulfill the UX recommendation of hiding blockchain intricacies. The system now handles all Cardano blockchain interactions server-side, so users can operate the platform without needing to manage keys or **ADA** tokens directly.

- **UI/UX Refinements:** Overhauling navigation and screen layouts to be more intuitive. This involved simplifying menus, improving page load times, and creating role-based dashboard views for each actor. For example, verifiers now see a tailored interface with their queue of pending verifications and clear action prompts, while project developers see monitoring status and data upload tools relevant to them.

- **Clearer Verification Process:** Updating the verification and approval workflow logic to be more transparent. The platform now includes step-by-step guidance for each verification stage and improved notification mechanisms (e.g. alerts when a verifier has approved or flagged an issue). These changes clarify how a credit moves from data submission to third-party validation and finally to issuance, addressing the earlier confusion.

This integration summary ensures that every critical piece of feedback was not only acknowledged but translated into concrete solution updates, resulting in a more user-friendly and robust platform.

5.4 Pilot-Ready Solution

A stable, functional version of Kijani MRV that is technically sound, policy-aligned, and ready for pilot deployment with selected carbon projects. The solution is designed to support real-world MRV operations while remaining adaptable for future regulatory integration.

By the end of the accelerator, the team delivers a stable, fully functional version of the **Kijani MRV** platform that meets both technical and policy requirements. This pilot-ready solution is built on the Cardano blockchain infrastructure and has been vetted for alignment with Ethiopia's emerging carbon market regulations and MRV guidelines. It is ready for deployment in real project environments (specifically the chosen pilot sites under the FOLUR and eQub initiatives) and capable of supporting on-the-ground MRV operations. The system's architecture remains flexible and modular, allowing for future regulatory integration or scaling to additional project types. Critically, the pilot solution adheres to best practices in digital carbon accounting leveraging blockchain's transparency and immutability to ensure trust in data integrity (as seen in similar Cardano-based sustainability applications). In sum, the output of Phase 5 is a solution that is technically sound, policy-aligned, and immediately usable in a field pilot setting.

5.5 Confirmed Pilot Partnership(s)

The pilot has initiated early stakeholder engagement to validate feasibility, secure institutional alignment, and confirm operational readiness for implementation. At this stage, partnerships are **not yet formalized** through signed MoUs; rather, they are in **preliminary and active scoping** status pending formal UNDP coordination and written requests.

Government and Policy-Level Engagement (In-Progress; not yet confirmed)

Preliminary engagement has been initiated with the **Ministry of Planning and Development (MoPD)** to assess strategic interest and policy alignment. MoPD has indicated that it would consider participating and providing coordination support **upon receipt of a formal UNDP request and defined collaboration scope** (including pilot objectives, governance structure, and roles). Accordingly, MoPD is treated as a **prospective strategic partner** at this stage, with confirmation dependent on formal process and internal approvals.

Anchor Use-Case Partners (Preliminary confirmation of availability; scoping underway)

The pilot has conducted **preliminary conversations with the FOLUR program** to assess interest and suitability as a pilot implementation pathway (particularly for forestry/landscape restoration MRV). Discussions to date have focused on the potential operational fit (data availability, field processes, and community readiness), and an initial group has **confirmed availability and willingness to participate** in pilot activities, subject to coordination and field scheduling.

In parallel, **eQub clean cookstove project** is positioned as the **primary pilot pathway** (as agreed), with initial internal planning underway to align MRV requirements, data collection workflow, and stakeholder participation parameters.

International Standards Bodies (Outreach planned; not yet confirmed)

Engagement with international standards and integrity bodies (e.g., **Verra, Gold Standard, Plan Vivo, IC-VCM**) is being **initiated through outreach**, but no confirmations have been secured to date. The intention is to align the pilot's MRV outputs and credit lifecycle design with recognized methodologies and integrity principles, and to seek technical guidance on how the pilot's evidence

model can be structured to meet market expectations. Formal engagement will depend on introductions and coordination supported by UNDP.

Solution Maker / Technology Delivery Partner (Confirmed)

Kitovu Technology is confirmed as the **Solution Maker** responsible for the technical design and delivery of the platform's architecture, including the blockchain-enabled registry and MRV workflow components (Cardano-based smart contracts, evidence anchoring, and buyer-facing proof views). The Solution Maker has committed development capacity during the Accelerator phase under a pro bono or discounted arrangement, with detailed implementation timelines captured under Section 4 and the post-acceleration roadmap.

5.6 Stakeholder Alignment Matrix

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement (Lead / Support / Advisory)	Engagement Modality (e.g., MOU, Contract)	Focal Point / Contact Person
UNDP Ethiopia (CRES Unit & relevant PMU)	Challenge Owner / Convenor	Overall coordination; technical oversight; alignment with UNDP climate, restoration, and digital innovation portfolios; facilitation of partner and government engagement	Lead	Internal UNDP coordination	Confirmed (internal UNDP role)
Equb (Clean Cooking Initiative)	Primary Pilot Implementation Partner (Clean Cooking)	Operational coordination for clean cooking MRV use case; facilitation of beneficiary participation; testing of household-level MRV workflows; feedback on usability	Lead (pilot use case)	Pilot coordination agreement	Availability to participate confirmed
Participating Communities / Household Groups	Local Pilot Participants	Participate in MRV data collection; provide field evidence (cookstove installation and usage); test mobile data capture workflows	Support	Community consent / participation agreement	Interest to participate pre-assessed

Kitovu Technology (Solution Maker)	Technical Development Partner	Design and develop blockchain-based MRV registry; deploy Cardano smart contracts; integrate evidence anchoring, dashboards, and proof-of-retirement views	Lead (technical)	Service agreement	Confirmed Solution Maker
Ministry of Planning and Development (MoPD)	Prospective Strategic & Policy Partner	Potential policy alignment; facilitation of inter-ministerial coordination upon formal UNDP request	Advisory	Formal UNDP request → potential MoU	Preliminary discussions held; engagement subject to formal UNDP request
Environmental Protection Authority (EPA)	Prospective Regulatory Stakeholder	Technical input on MRV integrity; observer role during pilot; feedback on regulatory and methodological alignment	Advisory	Policy consultation	To be engaged; not yet confirmed
FOLUR Programme / Coffee Cooperatives	Proposed Secondary Use Case (Agroforestry)	Provide landscape context; support assessment of agroforestry MRV applicability; potential future pilot expansion	Support / Advisory	Internal coordination	Proposed; not yet confirmed

Independent Verifier(s)	Third-Party Verification & Review	Review MRV workflows; assess verification logic; provide recommendations for audit readiness	Advisory / Support	Short-term consultancy	To be engaged during pilot
Global Standards Bodies (Verra, Gold Standard, Plan Vivo, IC-VCM)	Integrity Reference & Guidance	Non-binding technical guidance on alignment with international integrity principles	Advisory	Technical outreach	Outreach planned; no confirmation claimed
Digital & Financial Partners (Telebirr, M-Pesa, Kacha)	Payment and financial integration partner	Enable secure mobile money disbursements and integrate payout systems with the registry	Support	Technical MoU	Partnership Managers

5.7 Budget Proposal (Pilot)

The proposed pilot budget is structured in two phases to reflect both the scope of the SDG Blockchain Accelerator and the practical requirements of piloting a blockchain-enabled MRV system within Ethiopian context.

- **Phase I (Months 1–6)** focuses on core system development, testing, and demonstration aligned with the Accelerator program deliverables.
- **Phase II (Months 7–12)** represents a continuation budget required to sustain pilot operations, field validation, policy engagement, and iterative refinement beyond the Accelerator period. Phase II costs are indicative and subject to funding availability and pilot performance.

This phased structure avoids over-commitment while demonstrating a realistic pathway to a full 12-month pilot.

5.7.1 Phase I – Core Build Budget (Months 1–6)

Budget Line Item	Unit / Description	Quantity	Unit Cost (USD)	Total Cost (USD)	Funding Source
Cardano Blockchain Developer	Smart contracts, registry logic, DID integration	6 months	2,000	12,000	Pending
Backend Developer	MRV pipeline, APIs, registry workflows	6 months	1,500	9,000	Pending
Frontend Developer	Dashboards, onboarding, verifier & buyer views	6 months	1,000	6,000	Pending
Cloud & Infrastructure	Hosting, APIs, monitoring	6 months	1,000	6,000	Pending
Satellite Data Access	Pilot imagery & indices (NDVI/NDCI)	6 months	1,000	6,000	Pending
AI/ML Model Development	Carbon estimation, anomaly detection	6 months	2,000	12,000	Pending
IPFS / File Storage	Evidence storage and retrieval	6 months	500	3,000	Pending
UX Testing & Iteration	Usability testing and refinements	Lump sum	—	4,000	Pending
Policy & Stakeholder Clinics	Technical alignment sessions (EPA, UNDP)	Lump sum	—	4,000	Pending

Total Phase I (Core Build) Budget: <u>USD 62,000</u>

5.7.2 Phase II – Pilot Continuation Budget (Months 7–12, Indicative)

Phase II supports sustained pilot operation once the core system is built and demonstrated.

Budget Line Item	Unit / Description	Quantity	Estimated Cost (USD)	Notes
Continued Technical Support	Backend & blockchain maintenance	6 months	8,000	Reduced intensity
Satellite Data & Analytics	Ongoing monitoring	6 months	6,000	As needed
Field Enumerators & Local Support	Data collection, training (Equb pilot)	Lump sum	10,000	Pilot dependent
Verifier Engagements	Third-party review, workflow testing	Lump sum	6,000	Limited scope
Legal & Compliance Advisory	Carbon, data, sandbox compliance	Lump sum	6,000	As required
Travel & Logistics	Site visits, coordination	Lump sum	6,000	Conservative

Indicative Phase II Budget: <u>USD 52,000</u>

5.7.3 Summary Budget View

Budget Phase	Duration	Estimated Cost (USD)
Phase I – Core Build (Accelerator-aligned)	Months 1–6	~62,000
Phase II – Pilot Continuation	Months 7–12	~52,000

5.8 Impact Measurement Framework

5.8.1 Contribution to relevant SDGs and indicators

SDG 13 – Climate Action

- *Target 13.2 / Indicator 13.2.1*
- **Contribution:** Establishes a **pilot-grade, Article 6-aligned digital MRV and registry layer**, enabling Ethiopia to test transparent reporting and credit lifecycle tracking under real conditions.

SDG 15 – Life on Land

- *Target 15.2 / Indicator 15.2.1*

- **Contribution:** Enables **verified monitoring of agroforestry and restoration activities**, with conservative pilot coverage focused on selected cooperatives rather than national-scale claims.

SDG 7 – Affordable and Clean Energy

- *Target 7.1 / Indicator 7.1.2*
- **Contribution:** Supports **MRV readiness for improved cookstove programs**, beginning with Equb-led distribution pilots and laying groundwork for future scaling.

Cross-cutting (conservative)

- Tests pathways for mobilizing results-based climate finance.
- Strengthens digital MRV capacity among local implementers.
- Improves transparency and buyer confidence without inflating volumes.

5.8.2 Key Performance Indicators (KPIs)

To measure the pilot's success and impact, the following Key Performance Indicators have been defined. Each KPI includes a method of measurement, a target to be achieved over the 12-month pilot, and notes on its significance:

KPI	***Measurement Method***	***Target / Threshold***	***Notes***
Hectares of coffee landscapes restored and managed	Remote sensing analysis + field verification	≥ 2,000 hectares during pilot	Represents a subset of the FOLUR target areas
Carbon credits issued, transferred, and retired	Blockchain registry transaction records	≥ 5,000 tCO_2e (test lots) *Successful end-to-end issuance, transfer, and retirement of pilot-generated credits (at least one full cycle per project)*	Demonstrates full lifecycle functionality
Time from evidence submission to credit issuance	Measurement: System process analytics	≤ 30 days	Indicates efficiency gains compared to traditional MRV processes (benchmarking the speed of verification and issuance).

Value of performance-based payments disbursed	Payment logs from integrated mobile money systems	≥ 2 disbursement tranches per cooperative	Confirms that financial incentives reach local stakeholders (cooperatives) in a timely manner.
Adoption of digital MRV tools by cooperatives	User activity reports and training completion data	≥ 80% of participating cooperatives	Reflects the solution's usability and the capacity built among local organizations.
Compliance with global MRV and crediting standards	Verification by standards bodies (Verra, Gold Standard)	100% compliance for issued credits	Ensures international credibility and alignment with best-practice standards.

5.8.3 Monitoring & Reporting Plan

A clear monitoring and reporting plan will ensure that the pilot's progress is continuously tracked and communicated to stakeholders. Key elements of the plan include:

- **Continuous (Real-Time) Monitoring:** Automated, real-time data collection is enabled via satellite imagery, IoT sensors (where available), and digital field submissions. This provides instant visibility into project activities (e.g. tree planting progress, cookstove usage) as data is generated.

- **Monthly Updates:** Every month, the project team will compile and review aggregated performance metrics and system health data. These monthly check-ins include summary dashboards of key KPIs (credits generated, hectares monitored, etc.) and help in promptly addressing any operational issues.

- **Quarterly Reporting:** Comprehensive reports will be prepared on a quarterly basis, aligning with verification cycles of the carbon projects. Each quarterly report will be "verification-ready," meaning it packages the necessary data and documentation according to the methodologies and standards used, facilitating periodic review by verifiers or regulatory bodies.

- **Reporting Mechanisms**: A user-friendly online dashboard will be accessible to key stakeholders (UNDP, government partners, and cooperatives) for real-time data viewing. In addition, brief monthly status emails or calls will keep all partners informed. The quarterly reports will be formally submitted to the Challenge Owner (UNDP Ethiopia) and relevant government agencies (e.g. EPA), and can be shared with funding partners to demonstrate progress. The final pilot results will be documented in a comprehensive report at the end of 12 months, ensuring transparency and accountability.

Frequency

- ***Continuous (Real-Time):***
  Automated monitoring via satellite data, IoT inputs (where applicable), and digital field submissions.
- ***Monthly:***
  Aggregated project performance updates and system health monitoring.
- ***Quarterly:***
  Verification-ready reporting cycles aligned with project methodologies.
- ***On-Demand:***
  Reports generated for verifiers, regulators, or buyers as required.

Responsible Parties

- ***Project Developers:***
  Primary responsibility for submitting field data, activity logs, geotagged evidence, and periodic reports.
- ***Independent Verifiers:***
  Review submitted data, conduct validation checks, approve or request clarifications, and certify emission reductions.
- ***Program Management Unit (PMU) / Regulators:***
  Oversight of compliance, audit access, authorization of issuance, and alignment with national MRV and Article 6 requirements.
- ***Kitovu Technology Company (Platform Operator):***
  Maintains platform operations, data integrity, analytics, and reporting infrastructure.

Format

- ***Digital Dashboards:***
  Real-time visual summaries of project performance, emissions reductions, and verification status.
- ***Standardized MRV Reports:***
  Exportable PDF and spreadsheet formats aligned with national and international MRV standards.
- ***Blockchain-Backed Records:***
  Cryptographic proof-of-records (hashes, timestamps, signatures) anchored on-chain for auditability.
- ***Public Proof Pages:***
  Read-only verification and retirement summaries are accessible to buyers and stakeholders.

Tools

- ***Remote Sensing & AI/ML:***
  Satellite imagery and machine learning models for land-use, canopy cover, and activity verification.
- ***Mobile Data Capture:***
  Field-level data collection apps with geotagging and timestamping.
  Blockchain Infrastructure:
  Cardano smart contracts for issuance, transfer, and retirement; on-chain hashing of MRV dossiers.
- ***Decentralized Identity (DID):***
  Role-based access and authorization for developers, verifiers, and regulators.
  Geospatial & Analytics Platforms:
  Interactive maps, dashboards, and reporting tools for decision-making and transparency.

5.8.4 Risks Assessment & Mitigation Plan

Risk area	Specific risk (proposal context)	Likelihood	Impact	Mitigation measures	Owner	Monitoring KPI / trigger	Timeframe
Policy & regulation	Delays or changes in carbon market regulations slow the approval of registry/MRV pilots, as well as trading permissions.	Medium	High	Early and continuous engagement with EPA/MoPD; co-drafting of pilot protocols in regulatory sandbox; aligning artifacts with Ethiopia's Carbon Market Regulation and Digital Ethiopia strategy.	EPA, MoPD, UNDP	MoUs signed; # of joint policy workshops; sandbox admission obtained.	Preparatory phase (pre-pilot) and ongoing throughout the pilot.
Institutional coordination	Fragmented mandates (EPA, MoA, MoF, ECMA, NBE) cause delays and duplications.	Medium	High	Establish an Inter-ministerial Carbon Market Coordination Task Force with TORs, RACI for approvals, shared workplan.	MoF/EPA (co-chairs)	Task Force constituted; % of approvals meeting SLA.	Formation in Months 1–3 of the pilot, then ongoing oversight.
Land & carbon rights	Ambiguity over carbon rights under state land ownership undermines benefit-sharing and investor confidence.	Medium	High	Model legal clauses for carbon rights/leases; standardized community contracts; grievance & redress mechanism embedded.	MoA, MoJ, EPA	# projects using standard agreements; # grievances resolved within SLA.	Short to mid-term (initial legal frameworks in place early, with ongoing refinement)
Market integrity (double counting)	Double-counting or duplicate issuance across systems damages credibility.	Low	High	Canonical on-chain registry with unique IDs; retirement proofs; third-party verification; public audit trail.	EPA (registry), independent verifiers	% credits with on-chain retirement proof; zero duplicate events.	Build & steady-state
Article 6 transition	Uncertain Article 6.2/6.4 alignment and slow authorization processes limit international trades.	Medium	High	Produce Article 6 readiness module (authorization, CA ledger mock-up); pursue early bilateral pilots; align with UNFCCC guidance.	EPA, MoF, MoFA	Article 6 SOP approved; # of bilateral LOIs/MoUs.	Months 2–9 of the pilot (to develop and test Article 6 components) .
Verification cost & capacity	High MRV transaction costs exclude small developers; limited local verifier capacity.	Medium	Medium –High	Deploy AI/IoT-enabled MRV to lower costs; create a Carbon Project Incubator Fund for early-stage MRV; train local auditors (universities/NGOs).	UNDP, EPA, academia	Average MRV cost/credit; # local verifiers accredited.	Pilot & scale
Technology interoperability	Interoperability challenges across chains/systems stall integration.	Medium	High	Open standards; phased sandbox pilots; cross-chain anchoring (state roots); robust APIs.	Tech vendor, NBE Sandbox	# successful end-to-end tests; API uptime ≥99.5%.	Months 3–8

Cybersecurity & data privacy	Breach, key loss, or PII exposure (field data, DID).	Medium	High	Security by design; HSM/secure key custody; privacy-preserving storage (hash on-chain, evidence off-chain/IPFS); DPIA; incident response playbook.	Tech vendor, EPA (registry)	# critical vulnerabilities fixed; incident MTTR; audit pass rate.	Ongoing
Buyer demand	Lower-than-expected international demand or price volatility for Ethiopian credits.	Low–Medium	High	Engage ACMI & UNDP buyer networks; pre-marketing; diversify methodologies (cookstoves, forestry, soil); quality upgrades to meet IC-VCM.	UNDP, EPA	# of offtake MOUs; average price/ton; % inventory sold.	Months 4–12
Domestic adoption	Low uptake by cooperatives/SMEs due to awareness and skills gaps.	Medium	Medium	Targeted capacity building; incentives for early adopters; human-centered UX and local language support.	UNDP, CRES, Coop agencies	# active local projects onboarded; training completion rates.	Ongoing
Payments & FX	Constraints on cross-border settlements; mobile money integration hurdles.	Medium	Medium	Work via NBE/ECMA sandbox for tokenized escrow; integrate Telebirr/M-Pesa/Kacha for last-mile payouts; comply with forex rules.	NBE, ECMA, PSPs	Time to settle; % payouts via mobile money; sandbox test cases passed.	Months 3–9
Social & conflict sensitivity	Project siting in fragile areas risks tensions or exclusion; DDR linkages may stall.	Medium	High	Conflict-sensitive design; inclusive targeting; community monitoring nodes; link reintegration payments to verifiable milestones via smart contracts.	UNDP DDR/CRES, local govts	# grievances; social cohesion score; % DDR beneficiaries with verified earnings.	Pilot & scale
Environmental & safeguards	ESG non-compliance (biodiversity, FPIC, gender) jeopardizes certification.	Low–Medium	High	Apply Verra/GS safeguards; FPIC for community projects; gender-responsive benefit-sharing; transparent disclosure on registry.	EPA, Verifiers, Developers	# safeguard NCs; % projects with FPIC records; gender-disaggregated payouts.	Ongoing
Additionality	Credits not truly additional reduce integrity and price.	Medium	High	Conservative baselines; standardized additionality tests; third-party review before issuance.	Verifiers, EPA	% methodologies with standardized tests; rejection rate at validation.	Pre-issuance
Leakage	Activity displacement causes unaccounted emissions elsewhere.	Medium	Medium	Leakage buffers; regional landscape planning; monitoring of adjacent areas via remote sensing.	Developers, EPA	Leakage ratio; buffer drawdown events.	Monitoring phase

Permanence	Reversal risk (fire, drought, encroachment) undermines long-term climate value.	Medium	High	Buffer pools; long-term stewardship contracts; parametric insurance options; continuous satellite monitoring & alerts.	EPA, Developers	Buffer coverage (%); # reversal incidents detected and compensated.	Post-issuance
Reputational	Public criticism of carbon credits/"greenwashing" affects offtake.	Medium	Medium –High	Proactive transparency (open data room, third-party reviews); align to IC-VCM Core Carbon Principles; clear claims guidance for buyers.	UNDP, EPA	# public disclosures; buyer satisfaction/NPS; media risk rating.	Ongoing
Delivery schedule	Slippage on MVP milestones (registry, MRV, DID integration).	Medium	Medium	Agile delivery with gated reviews; critical-path tracking; contingency resourcing; demo-ready pilots by Week 12 plan.	Tech vendor, UNDP AccLab	% milestones on time; burn-down charts; number of Sev-1 bugs at demo.	Weeks 1–12
Financing	Insufficient co-financing for scale-up beyond pilot.	Medium	High	Blend BIOFIN/GEF/NORAD resources; link to ESX green bonds/DBE instruments; structure results-based payments.	UNDP IFL, MoF, DBE	$ leveraged; # instruments launched; % pilot costs covered.	Months 4–12
Domestic policy shifts	Macroeconomic or governance shifts affect the enabling environment (payments, FX, permits).	Medium	Medium –High	Scenario planning, policy risk register; maintain multi-stakeholder governance, so the platform remains policy-agnostic.	UNDP, EPA, NBE	Quarterly risk review; # policy changes requiring platform updates.	Quarterly

5.9 Post-Acceleration Roadmap & Sustainability Plan

After the acceleration program, the focus shifts to implementing the full 12-month pilot and setting the foundation for long-term sustainability. This Post-Acceleration Roadmap and Sustainability Plan outlines how the solution will be scaled and sustained through distinct phases, a dedicated budget, revenue models, strategic partnerships, and a clear fundraising strategy for future growth.

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)	**Medium Term (6–18 months)**	**Long Term (18+ months)**
Objective Deliver and validate a functioning digital MRV	**Objective**	**Objective** Institutionalize Kijani MRV as a trusted digital public

verification layer through real pilot use cases **Key Milestones** Deploy MVP of Kijani MRV platform (registry core + MRV pipeline). Complete field-based pilots in **Jimma (FOLUR coffee landscapes)** and **Bensa (eQub clean cooking)**. Integrate mobile field data capture (offline-first) for enumerators and cooperatives. Conduct controlled **issuance → transfer → retirement simulations** on Cardano testnet. Implement backend-managed wallet infrastructure (no end-user crypto interaction). Enable buyer-facing read-only provenance and retirement proof pages. Complete usability testing and UAT with cooperatives, field agents, and UNDP teams. Hold policy validation clinics with **EPA, MoPD, and UNDP CRES/FOLUR**.	Expand adoption, strengthen governance integration, and prepare for market transactions. **Key Milestones** Expand MRV coverage to additional cooperatives and clean cooking operators. Upgrade the platform for **multi-project and multi-region support**. Introduce **production-grade registry deployment** (mainnet-ready, subject to approvals). Formalize interoperability with national MRV and registry systems via APIs. Onboard accredited third-party verifiers and auditors. Implement automated MRV analytics (satellite + ML) at landscape scale. Launch transaction-based revenue model (subscriptions + issuance fees). Initiate Article 6 readiness modules (authorization workflows, CA ledger mockups).	good supporting climate finance at scale. **Key Milestones** Support national-level MRV and carbon registry operations (licensed deployments). Integrate additional methodologies (soil carbon, livestock, bioenergy). Enable cross-border Article 6 cooperation and ITMO tracking. Embed platform into national climate finance instruments (green bonds, RBP). Expand to other African markets through UNDP and government partnerships. Achieve financial sustainability through diversified revenue streams. Establish a regional MRV innovation hub and local capacity-building programs. **Outputs** Nationally embedded MRV infrastructure

Deliver the final Accelerator demo and documentation. **Outputs** Pilot-ready MVP Verified MRV datasets Testnet-issued and retired credit lots Policy-aligned technical documentation	Secure first commercial offtake agreements. **Outputs** Multi-project operational platform Revenue-generating registry Verified, market-ready credits Government-aligned integration pathways	Scalable climate finance pipeline Long-term operational sustainability

5.9.2 Budget for Phased Implementation (post-acceleration)

Budget Line Item	**Description**	**Estimated Cost (USD)**
Core Platform Engineering & Scaling	**Backend hardening, performance optimization, multi-project support, and role-based access**	**$210,000**
Smart Contracts & Registry Logic	**Cardano mainnet readiness, issuance/retirement logic, audits, monitoring**	**$120,000**
MRV Analytics & AI Models	**Satellite analytics, ML calibration, landscape-scale monitoring**	**$180,000**

Mobile Data Capture & UX	**Offline functionality, localization, and enumerator workflows**	**$90,000**
Infrastructure & Storage	**Cloud hosting, IPFS/Filecoin, monitoring & backups**	**$110,000**
Payments & Settlement Integration	**Mobile money integration, escrow logic, compliance**	**$95,000**
Governance & Interoperability	**APIs Integration with national MRV, reporting exports, and policy alignment**	**$85,000**
Field Deployment & Capacity Building	**Enumerator training, field operations, and user onboarding**	**$140,000**
Security, Privacy & Compliance	**Penetration testing, key management, and data protection**	**$75,000**
Operations, Support & Maintenance	**DevOps, monitoring, bug fixes, upgrades (12 months)**	**$120,000**
Independent Audits & Validation	**Technical, MRV, and financial audits**	**$80,000**

Total Estimated	**Post-Pilot Budget**	**$1,305,000**

- **Targeted Funding Sources:** To finance this budget, a mix of grants and co-financing will be pursued. Potential sources include UNDP's BIOFIN program, the Global Environment Facility (GEF), Norway's NORAD, Green Climate Fund (GCF) readiness grants, and bilateral donors (e.g. EU, Germany, etc.). The team will also explore private sector contributions, such as carbon credit **offtake** pre-financing from companies interested in purchasing the credits generated by the pilot.

- **Future Revenue Streams:** In the longer term, sustaining the platform will rely on generating revenue. The model is expected to include small transaction fees on credit issuances or trades, subscription fees for enterprise users (e.g. organizations that use the platform extensively or integrate it into their systems), and fees for premium registry services (such as analytics or reporting services for project developers). These revenues are projected to offset operational costs after the initial grant-funded pilot phase.

5.9.3 Financial Sustainability Model

A multi-pronged financial sustainability model will ensure that the platform can be maintained and scaled beyond the initial pilot phase. The model includes revenue generation mechanisms and a timeline for achieving cost recovery:

- **Transaction Fees on Credit Issuance & Transfers (Year 1–2):** The platform will levy a small fee per carbon credit (or per batch of credits) at the point of issuance and/or transfer to a buyer. This ensures that revenue is directly tied to market activity. The fee will be kept minimal to remain significantly lower than traditional registry fees, thereby encouraging projects to use the platform while still contributing to maintenance costs.

- **Verification & MRV-as-a-Service (Year 2+):** As the platform matures, Kijani MRV can offer *MRV-as-a-Service* to projects or programs that lack in-house capacity for advanced MRV. This service could include AI-assisted MRV calculations, satellite monitoring and reporting, and end-to-end verification workflow management. It would be provided for a service fee (e.g. priced per project or per hectare monitored). This model is especially attractive for forestry, agroforestry, and cookstove programs that need robust MRV without investing in their own systems.

- **Data & Insights Services (Optional, Year 4+):** In later stages, aggregated and anonymized data insights could be packaged for interested parties (e.g. climate funds, researchers, or policymakers). For example, the platform could provide sector-level trends, benchmarks on project performance, or impact analytics. Only non-sensitive, consent-based data would be shared, ensuring privacy. If pursued, this would create an additional revenue stream through subscriptions or one-off reports for those who value such insights.

- **Cost Recovery and Profitability Outlook:**

- The pilot phase (Year 1) is expected to be **grant-funded and focused on learning**, so revenue generation will be limited in the first year as the emphasis is on demonstrating the solution and impact.

- **Break-even is targeted within 24–30 months** after the pilot period. By the end of Year 2 or early Year 3, the combination of transaction fees and service offerings should cover operational costs if adoption and market activity grow as projected.

- The platform's design ensures a **low marginal cost** for adding each new project or user. Being cloud-based and modular, once the core infrastructure is in place, onboarding additional projects entails minimal incremental expense, improving profitability as scale increases.

- **Long-term sustainability** will hinge on recurring revenues (such as annual subscriptions by large project developers or government agencies) supplemented by transaction-based income as carbon market volumes increase. As African and global voluntary carbon markets scale, the platform's revenue could grow substantially, creating a self-sustaining model.

5.9.4 Partnership and Co-Financing Commitments

The success and scale-up of the Kijani MRV platform depend on a broad coalition of partners bringing technical expertise, policy support, and financing. Key partnership commitments include:

- **Government and Regulatory Partners:**

  - *Environmental Protection Authority (EPA)* – **Role:** Provides regulatory oversight, ensuring the pilot aligns with national carbon market governance and MRV standards. **Contribution:** Facilitates policy alignment by reviewing/approving registry protocols and integrating Kijani MRV with Ethiopia's official Carbon Market Regulation framework.

  - *Ministry of Planning and Development (MoPD) & Ministry of Finance (MoF)* – **Role:** Ensure strategic coordination across government initiatives and explore fiscal incentives for carbon projects. **Contribution:** Offer institutional backing for Article 6 readiness (international trading) and help link the pilot with national climate finance strategies (e.g. integrating results into development planning or budgeting).

  - *National Bank of Ethiopia (NBE) & Ethiopian Capital Market Authority (ECMA)* – **Role:** Oversee financial market integration and provide a regulatory sandbox for fintech innovations. **Contribution:** Support testing of the tokenized carbon credit transactions and escrow mechanisms under controlled conditions, and advise on compliance with financial regulations for carbon trading and cross-border payments.

- **Technical Partners:**

  - *Kitovu Technology Company (Nigeria)* – **Role:** Lead technology developer for the solution. **Contribution:** Designs, develops, and deploys the blockchain infrastructure, smart contracts, and MRV integration components of Kijani. Also implements decentralized identity (DID) systems for secure participant identification.

  - *Local Innovation Hubs (e.g. IceAddis, Orbit Innovation)* – **Role:** Provide incubation support and local tech ecosystem integration. **Contribution:** Offer workspace,

mentorship, and user-centric design expertise. They will help adapt the solution to the local context, organize user testing sessions, and build capacity among young developers and users in Ethiopia.

- **Financial and Development Partners:**
    - *UNDP Ethiopia (Innovative Finance Lab & Accelerator Lab)* – **Role:** Acts as a convenor and technical advisor, while also co-financing aspects of the pilot. **Contribution:** Leverages UNDP's BIOFIN (biodiversity finance) and PISTA (PrototypeLab for Sustainable Transition) initiatives to provide seed funding and ensure the project is embedded in Ethiopia's broader sustainable finance efforts. UNDP also assists with policy dialogues and stakeholder coordination.
    - *Global Environment Facility (GEF) & Norwegian Agency for Development Cooperation (NORAD)* – **Role:** International funding partners providing grants. **Contribution:** GEF and NORAD have indicated interest in financing the pilot implementation and associated capacity building (such as training local communities and officials in MRV), thereby covering a portion of the pilot's budget requirements.
    - *World Bank & FSD Africa* – **Role:** Knowledge and network partners focusing on financial innovation. **Contribution:** The World Bank (through programs like Ci-Dev or others) and FSD Africa (a development finance catalyst) will provide advisory support on structuring innovative finance instruments (e.g. green bonds, carbon-linked loans). They may also assist in mobilizing additional funds for scaling the platform after a successful pilot.
- **Project Developers:**
    - *FOLUR & DDR Projects (UNDP Ethiopia initiatives)* – **Role:** Serve as anchor pilot projects supplying carbon credits. **Contribution:** The UNDP-led FOLUR (Food Systems, Land Use and Restoration) program in Jimma will generate forestry/agroforestry credits, and the DDR (Disarmament, Demobilization and Reintegration) program's clean cooking initiative in Bensa will generate cookstove credits. These credits will be among the first registered and tracked on Kijani, demonstrating its functionality.
    - *World Vision Ethiopia, EthioTrees, C-Quest Capital* – **Role:** Implementing partners for community-level carbon projects. **Contribution:** World Vision and EthioTrees will bring additional reforestation and land restoration projects into the pilot, while C-Quest Capital contributes energy-efficient cookstove projects. Together, they supply a pipeline of high-integrity carbon credits and provide on-the-ground feedback on the platform's usability and impact.
- **Standards and Verification Bodies:**
    - *Verra, Gold Standard, Plan Vivo* – **Role:** Ensure the credits from the pilot meet international quality standards. **Contribution:** While not funding partners, these standard-setting organizations will engage with the pilot by reviewing methodologies and, if possible, accrediting the pilot-generated credits. Their involvement lends credibility, as credits verified to Verra's VCS, Gold Standard, or Plan Vivo methodologies can be more readily sold on international markets. They may also provide guidance to align Kijani's digital MRV approach with requirements for validation and verification under their standards.

- **(Potential) Co-Financing Commitments:**

    - *UNDP BIOFIN & PISTA:* UNDP's BIOFIN initiative (which identifies and mobilizes innovative finance for biodiversity/climate) and the PISTA program **have potential** to **commit seed funding** to kick-start platform development and support policy integration efforts (e.g. drafting necessary guidelines for digital MRV).

    - *GEF and NORAD:* The Global Environment Facility and NORAD have **the potential** to cover a significant portion of the pilot's implementation costs and MRV capacity-building activities at the local level.

    - *Private Sector & Carbon Buyers:* Several corporate climate leaders have expressed intent to enter into offtake agreements (e.g. forward purchases of credits). These potential buyers (including participants in initiatives like Microsoft's Climate Innovation Fund or Shopify's Sustainability Fund) could provide **pre-financing** that supplements grant funding, in return for a share of the credits that will be generated.

    - *Domestic Financial Institutions:* Local banks and the upcoming Ethiopian Securities Exchange (ESX) are involved in **co-financing through green investment instruments**. For example, an Ethiopian bank might allocate a green loan to support cooperatives in the pilot, or the ESX could facilitate a pilot green bond. These institutions are being engaged so that the pilot's success can translate into scalable finance (like green bonds or carbon credit-backed securities) in the domestic market.

5.9.5 Fundraising Plan

The fundraising strategy for scaling the blockchain-based carbon registry is designed to tap into diverse funding sources over time. It aligns with Ethiopia's Integrated Sustainable Financing Strategy and UNDP's Innovative Finance Lab framework, emphasizing a blend of public and private financing as well as reinvestment of carbon revenues. Key components of the plan include:

Targeted Funding Sources: The following categories of funding will be pursued to support the pilot and subsequent scale-up:

- **a) Multilateral and Bilateral Grants:** Multilateral climate finance sources and country donors are primary targets for initial funding.

    - *Global Environment Facility (GEF)* and *NORAD* – Provide co-financing grants earmarked for the pilot phase and building national MRV capacity (as noted in partnership commitments).

    - *Green Climate Fund (GCF)* – A proposal to GCF is planned to seek medium-term funding (for years 2–3) to support scaling the platform nationally and ensuring Article 6 readiness (including establishing the necessary authorization procedures and infrastructure for international trading).

    - *Bilateral Donors* (e.g. the EU, Germany, UK, Japan) – Engage through their climate and development programs for support in technology deployment and community-based project expansion. These donors often fund initiatives in Ethiopia around clean energy, forestry, and digital transformation, which aligns well with the project's outcomes.

- **b) Ecosystem and Innovation Funds:** Dedicated funds focused on innovative finance and market-building will be tapped.
    - *UNDP BIOFIN & PISTA* – As mentioned, UNDP's BIOFIN facility and its Prototype Fund (PISTA) have the potential to provide initial support; additional rounds of funding may be requested as the project shows results (especially if scaling to new regions or integrating new methodologies).
    - *African Carbon Markets Initiative (ACMI)* – This Africa-focused coalition, launched at COP27, offers both technical assistance and catalytic funding to increase the supply of African carbon credits. The project will leverage ACMI's support for expanding voluntary carbon market infrastructure in Ethiopia, potentially accessing grant funding or in-kind technical support[2].
- **c) Private Sector and Impact Investors:** The strategy recognizes the growing interest of private companies and impact investment funds in high-quality carbon credits and climate tech solutions.
    - *Corporate Climate Buyers* – Companies with carbon neutrality goals (for example, those participating in schemes like the Microsoft Climate Innovation Fund or voluntary buyers on platforms like Shopify's Sustainability Fund) will be approached for offtake agreements. Through such agreements, companies commit to buying a portion of the credits from the pilot upfront, providing early revenue (or even pre-payment) that can fund operations.
    - *Green Bond Investors via ESX* – As Ethiopia operationalizes its Securities Exchange, there is potential to issue **green bonds** or similar instruments that are linked to the carbon credits generated (for instance, bond proceeds finance the project and bond yield is partly tied to credit sales). Impact investors who purchase these bonds effectively channel capital into the project in return for financial and environmental returns. The fundraising plan includes collaboration with financial advisors to structure such blended finance instruments.
- **d) Carbon Market Revenues:** Finally, the project will aim to become self-sustaining by generating revenue directly from the carbon market
    - *Sale of Verified Credits* – Proceeds from selling the carbon credits produced by pilot projects (forestry, agroforestry, clean cooking) will be recycled into project funding. By demonstrating the market value of these credits, the pilot can attract more projects and investors.
    - *Tokenization and Retail Investment* – In the future, the platform may enable fractional ownership of credits (for example, tokenizing credits so that small investors or community members can own a share). This can open up a new stream of micro-investments and broaden the base of support, while also raising public awareness and engagement in climate action.

Fundraising Timeline

To operationalize the above strategies, a phased timeline for fundraising has been outlined:

- **Phase 1: Pilot Financing (Months 1–6):** Secure the initial seed funding and grants to launch the pilot. Additionally, technical support from ACMI will commence to bolster the pilot. By Month 6, the goal is to have the pilot fully funded and demonstrate early results as a basis for the next round of fundraising.

- **Phase 2: Market Readiness (Months 6–12):** As the pilot progresses, focus shifts to enabling market transactions. During this period, negotiations with corporate off-takers (buyers of credits) will be formalized, some buyers may agree to purchase credits as they are issued, providing revenue. The project will also begin onboarding impact investors, for example by presenting the pilot at investor forums. Integration with the ESX green bond platform is targeted by the end of this phase, potentially to prepare a bond or note issuance linked to the project's performance.

- **Phase 3: Scale-Up Financing (Year 2–3):** With a successful pilot (after 12 months) as proof of concept, a larger scale funding round will be pursued. In Year 2, a proposal to the GCF will be submitted, aiming for a significant grant or concessional investment to support national scale rollout (this process can take 6–12 months for approval). Bilateral donor engagement will be intensified, leveraging pilot results to secure new agreements. Meanwhile, the team will work on structuring blended finance instruments such as green bonds or carbon credit-backed loans – these might be launched in Year 3 in collaboration with the Ministry of Finance and international banks.

- **Phase 4: Sustainability (Year 3 onward):** By the third year and beyond, the platform is expected to start generating steady income from its operations. Carbon credit sales revenue will ramp up as more projects come online and existing projects mature (producing more credits annually). A portion of these revenues will be reinvested into improving the platform and expanding to new communities, while another portion could provide returns to investors or maintenance of operations without external aid. Essentially, after Year 3, the aim is for the initiative to rely less on grants and more on its own revenue-generation and impact investment recycling.

Resource Mobilization Targets

To monitor success, the following fundraising targets have been set for each stage:

- **Pilot Phase:** ~$0.5–$1 million in funding, primarily from UNDP (and its partners) and initial grants (GEF, NORAD), supplemented by any private co-financing (e.g. in-kind contributions or early buyer agreements).

- **Scale-Up (Post-pilot, into Year 2–3):** ~$3–$5 million to be raised for the national roll-out and multi-year scale. This is expected from a combination of GCF approval, additional bilateral donors, and innovative instruments like green bonds as they come to fruition.

- **Long-Term (By 2030):** Aim for the platform to facilitate ~$50 million per year in carbon finance flows by 2030. This figure represents the **annual carbon revenue** (from credit sales) that could be realized if the platform scales to support a significant portion of Ethiopia's carbon market (and potentially regional markets). Hitting this target would indicate a self-sustaining market-driven mechanism is in place, far beyond the pilot.

5.11 Legal & Compliance Considerations

Implementing a blockchain-based carbon registry in Ethiopia requires careful navigation of legal and regulatory frameworks. The solution must comply with national laws and international standards to ensure integrity and legitimacy. Key legal and compliance considerations include:

- **Carbon Market Regulation:** Ethiopia's emerging Carbon Market Regulation (2024) provides the legal foundation for operating both compliance and voluntary carbon markets in the country. The Kijani MRV platform will adhere to all requirements for credit issuance, transfer, and retirement as stipulated by this regulation. This ensures that the pilot remains

aligned with national MRV protocols and the provisions of Article 6 of the Paris Agreement (which governs international trading of credits). Any methodologies used will be vetted to comply with what the regulation recognizes.

- **Financial Sector Compliance:** Since the platform interfaces with payment systems and financial transactions (e.g. paying communities, trading credits of financial value), it must comply with Ethiopia's financial rules:
    - *National Payment System Proclamation (2020, amended 2023)* – governs electronic payments and requires that any payment service or instrument operate under licensing and oversight of the National Bank of Ethiopia. The project will either partner with licensed payment providers or operate within the NBE's regulatory sandbox to remain compliant.
    - *Open API Directive (2024)* – mandates interoperability among fintech platforms. Kijani's design is being built to open standards, and APIs will be provided/documented to ensure it can interoperate with banks and mobile money services in line with this directive.
    - *NBE Regulatory Sandbox Protocols* – The National Bank's sandbox allows controlled testing of innovations like tokenized credits and digital escrow. The pilot will function within these sandbox guidelines for any aspect involving digital tokens or cross-border transactions, ensuring any new financial products (e.g. a tokenized carbon credit) meet central bank requirements.

- **Data Privacy & Cybersecurity:** The platform will handle sensitive data (project details, personal data of cooperative members, digital IDs, etc.), so robust data protection measures are critical. Compliance steps include:
    - Adherence to Ethiopia's data protection guidelines (draft provisions that are in line with international best practices) to ensure personal and project data is collected and stored lawfully and with consent.
    - Implementing a privacy-preserving architecture – for example, only hashes or encrypted references of data are stored on-chain, while actual detailed data (names, exact coordinates, etc.) are kept off-chain in secure databases or IPFS with access controls. This prevents public exposure of sensitive information while leveraging blockchain transparency for proofs.
    - Following cybersecurity standards (such as ISO 27001 for information security) and having an incident response plan in place. Regular security audits will be conducted. In case of any data breach or unauthorized access attempt, there are defined protocols to contain and report incidents, in compliance with any national cybersecurity directives.

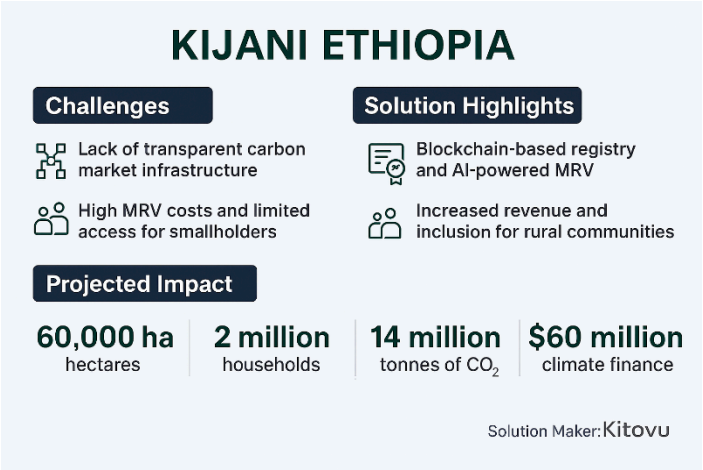
- **Digital Identity and KYC:** The solution plans to use Decentralized Identity (DID) for participants (projects, verifiers, buyers). Any digital ID system employed must align with Ethiopia's national digital identification framework and fulfill Know-Your-Customer (KYC) requirements for financial transactions. This means, for example, if community members are to receive payments via mobile money, their identities must be verified in line with telecom/banking KYC rules. The platform will integrate with the national ID (if available, e.g. Fayda ID system) to securely accredit and verify identities of users without exposing personal data.

- **International Standards and Certification:** Carbon credits produced will be structured to meet international quality standards. This involves aligning with the **IC-VCM Core Carbon Principles**, a set of high-integrity requirements defined by the global Integrity Council and using approved methodologies from bodies like Verra, Gold Standard, and Plan Vivo. Compliance here means the credits from Kijani MRV can be recognized and tradable on global markets, as they will carry the necessary stamps of approval (for instance, a Verra VCU or Gold Standard label) if the pilot project documents are submitted to those standards for verification.

- **Intellectual Property and Licensing:** The software developed (smart contracts, code for the registry and MRV algorithms) will respect open-source licenses as applicable. For instance, if the solution builds on existing open-source blockchain frameworks, it will continue to comply with those licenses (MIT, Apache 2.0, etc.). Any original code by Kitovu or partners will be appropriately licensed to the implementing partners or government to ensure long-term usability. Additionally, data generated by the project will be handled under agreements that clarify ownership (e.g. communities retain ownership of their project data, while summary statistics might be open for public good).

- **Anti-Money Laundering (AML) & Counter-Terrorism Financing (CTF):** The platform must diligently prevent illicit financial flows. All tokenized carbon credit transactions, especially if foreign buyers are involved, will be screened for compliance with Ethiopia's AML/CTF regulations and FATF (international) standards. This includes performing due diligence on buyers (to ensure they are legitimate entities), monitoring transactions for suspicious patterns, and reporting any large or unusual transactions to relevant authorities. By doing so, the project safeguards against risks that the carbon market platform could be misused for money laundering or terrorism financing, thereby protecting its integrity and stakeholders.

5.12 Visibility Assets / Communication Materials



To maximize the pilot's visibility and communicate its progress and successes to a wide audience, a set of communication materials will be developed. These assets will help in keeping stakeholders informed, attracting potential partners, and sharing learnings from the project:

- **Press Release (Pilot Launch):** A press release will announce the launch of the Kijani MRV pilot, highlighting the partnership between UNDP, government agencies, and Kitovu. It will outline the pilot's goals, the innovation of using blockchain for MRV, and the expected benefits for communities and climate action. This will be distributed to local and international media at project kick-off, and again at key milestones (e.g. upon successful completion of the pilot, highlighting results).

- **Project Brief / Fact Sheet:** A concise two-page brochure will be prepared, summarizing the challenge, the Kijani MRV solution, pilot design, and expected outcomes. This fact sheet can be shared with government officials, donors, and participants to quickly get up to speed on the project. It will use infographics and simple language to explain how the digital MRV platform works and how it ties into Ethiopia's SDG and climate targets.

- **Demo Video:** A short video (3–5 minutes) will be produced to visually demonstrate the platform in action. It will likely include a screen-recording of the software interface showing how a carbon credit is tracked, combined with footage from the field (e.g. farmers measuring trees or women using clean cookstoves, with data being uploaded). The video will be used in presentations (such as the final demo day) and shared on social media to generate interest among potential supporters.

- **Blog Posts and Case Study Articles:** Throughout the pilot, the team will publish periodic blog updates on UNDP's website or Medium, detailing progress, challenges, and lessons learned. One comprehensive case study article will be written after the pilot's completion, documenting the impact (e.g. hectares restored, credits issued, community benefits) and experiences. These stories will not only provide transparency but also help other countries or projects learn from Ethiopia's experience in digital MRV.
- **Social Media Content:** To engage a broader audience, a series of social media posts will be created (for platforms like Twitter, LinkedIn, and Facebook). These will include infographics summarizing key achievements (for example, "2,000+ hectares under improved MRV" or "First blockchain-registered carbon credit issued in Ethiopia!"), quotes from beneficiaries or officials, and short video snippets from the field. A social media toolkit will be provided to partners so that UNDP, government ministries, and Kitovu can consistently amplify the pilot's messages.

- **Visual Assets for Events:** In addition to digital content, the project will prepare presentation slides, banners, or posters that can be used in workshops, stakeholder meetings, or conferences. For instance, if the team presents at a climate finance forum, they will have a polished slide deck and perhaps a roll-up banner illustrating the project's concept and partners. These materials ensure the project is well-represented in public forums and maintains a professional image.

All visibility and communication materials will follow UNDP and donor branding guidelines, ensuring proper acknowledgement of all partners and alignment with the communications strategy of the SDG Accelerator.

5.13 Pitch Deck

The presentation is uploaded in our google drive folder.

5.14 Final Demo Presentation

Can be done upon request.